CHAPTER

19

Layout and Packaging

In the past 40 years, analog CMOS circuits have evolved from low-speed, low-complexity, small-signal, high-voltage topologies to high-speed, high-complexity, low-voltage "mixed-signal" systems containing a great deal of digital circuitry. While device scaling has enhanced the raw speed of transistors, unwanted interaction between different sections of integrated circuits as well as nonidealities in the layout and packaging are increasingly limiting both the speed and the precision of such systems. Today's analog circuit design is very heavily influenced by layout and packaging.

In this chapter, we study principles of layout and packaging, emphasizing the effects that manifest themselves when analog and digital circuits coexist on a chip. For the sake of brevity, we use the term "analog" to mean both "analog" and "mixed-signal." Beginning with an overview of layout design rules, we study a number of topics related to the layout of analog circuits, including multifinger transistors, symmetry, reference distribution, passive device layout, and interconnects. Next, we deal with the problem of substrate coupling. Finally, we describe packaging issues, analyzing the effect of self- and mutual inductance and capacitance of external connections to integrated circuits.

19.1 ■ General Layout Considerations

The layout of an integrated circuit defines the geometries that appear on the masks used in fabrication. From Chapter 18, the geometries include n-well, active, polysilicon, n^+ and p^+ implants, interlayer contact windows, and metal layers.

Figure 19.1 shows an example, where the mask geometries required for a PMOS transistor are drawn. It is important to note the following: (1) the n-well surrounds the device with enough margin to ensure that the transistor is contained in the well for all expected misalignments during fabrication; (2) each

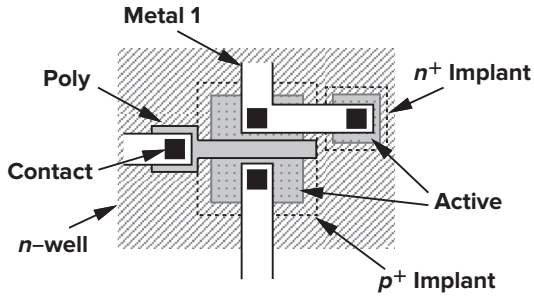


Figure 19.1 Layout of a PMOS transistor.

"active" area (S/D regions and n^+ contact to the well) is surrounded by a proper implant geometry with enough margin; (3) from the fabrication steps described in Chapter 18, the gate requires its own mask; (4) the contact windows mask provides connection from active and poly regions to the first layer of metal.

In most modern layout tools, the implants and even the n-wells are automatically generated from the remainder of the transistor geometries, reducing the number of layers that the layout designer draws or sees on the computer screen and simplifying the task.

19.1.1 Design Rules

While the width and length of each transistor are determined by circuit design, most of the other dimensions in a layout are dictated by "design rules," i.e., a set of rules that guarantees proper transistor and interconnect fabrication despite various tolerances in each step of processing. Most design rules can be categorized under one of the four groups described here.

Minimum Width The widths (and lengths) of the geometries defined on a mask must exceed a minimum value imposed by both lithography and the processing capabilities of the technology. For example, if a polysilicon rectangle is excessively narrow, then, owing to fabrication tolerances, it may simply break or at least suffer from a large local resistance (Fig. 19.2). In general, the thicker a layer, the greater its minimum allowable width, indicating that as technologies scale, the thickness must be decreased proportionally. Figure 19.3 depicts examples of minimum widths in a 40-nm technology. Note that the thickness of the layers is not under the control of the layout designer.



Figure 19.2 Excessive width variation in a narrow poly line.

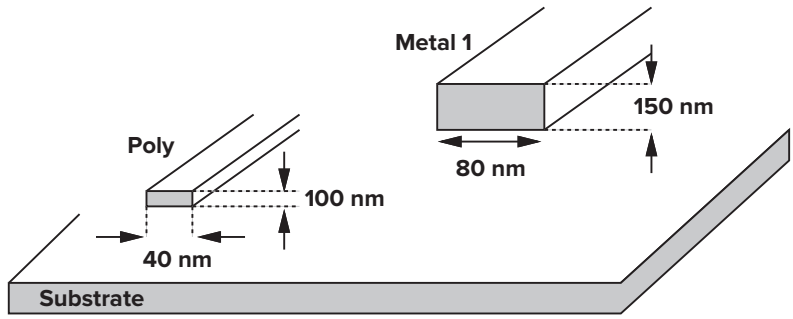


Figure 19.3 Widths and thicknesses of poly and metal lines.

Minimum Spacing The geometries built on the same mask or, in some cases, different masks must be separated by a minimum spacing. For example, as shown in Fig. 19.4(a), if two polysilicon lines are placed too close to each other, they may be shorted. As another example, consider the case shown in Fig. 19.4(b), where a polysilicon line runs close to the S/D area of a transistor. A minimum spacing is required here to ensure that the implant surrounding the transistor does not overlap with the poly line.

Minimum Enclosure We mentioned earlier that in the layout of Fig. 19.1, the n-well and the p^+ implant must surround the transistor with sufficient margin to guarantee that the device is contained by these geometries despite tolerances. These are examples of minimum enclosure rules. Figure 19.5 depicts

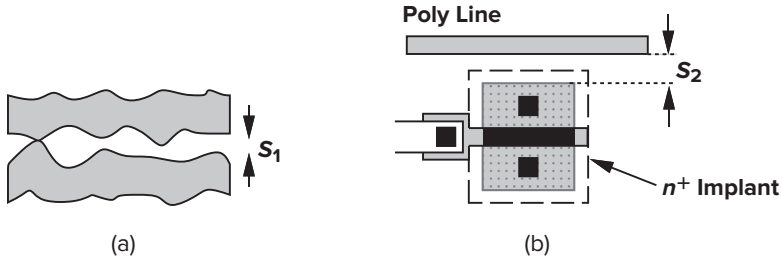


Figure 19.4 (a) Short between two excessively close poly lines; (b) minimum spacing between active and poly.

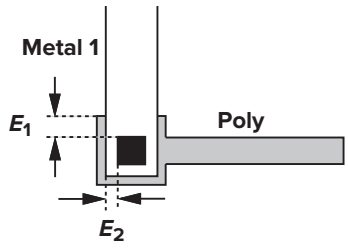


Figure 19.5 Enclosure rule for poly and metal surrounding a contact.

another example, where a poly contact window connects a poly line to a metal 1 line. To ensure that the contact remains inside the poly and metal 1 squares, both geometries must enclose the contact with enough margin.

Minimum Extension Some geometries must extend beyond the edge of others by a minimum value. For example, as shown in Fig. 19.6, the gate polysilicon must have a minimum extension beyond the active area to ensure proper transistor action at the edge.

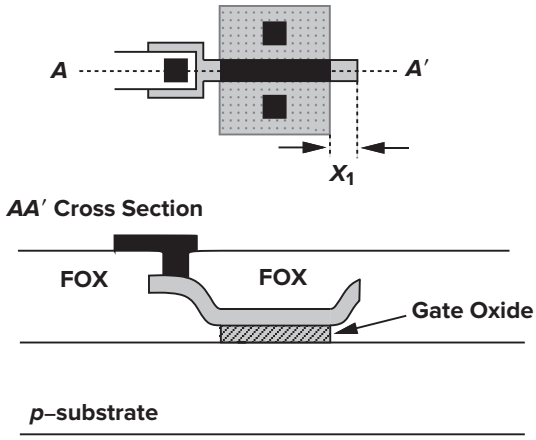
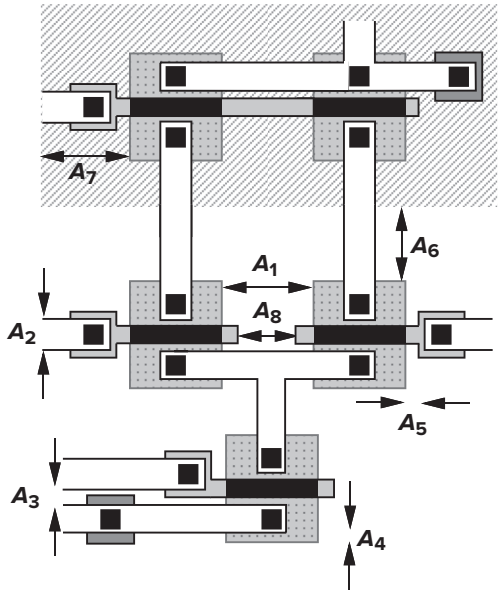


Figure 19.6 Extension of poly beyond the gate area.

In addition to the minimum dimensions specified in the previous four categories, some *maximum allowable* dimensions may also be enforced. For example, for long metal wires, the minimum width is typically larger than that for short wires to avoid "liftoff" problems. Other such rules relate to the "antenna effect," described in the next section.

Figure 19.7 summarizes a small subset of design rules governing the layout of an NMOS differential pair with PMOS current-source loads. Modern CMOS technologies typically involve several hundred layout design rules.



A_1 : Active–Active Spacing

A_2 : Metal Width

A_3 : Metal–Metal Spacing

A_4 : Enclosure of Contact by Active

A_5 : Poly–Active Spacing

A_6 : Active–Well Spacing

A_7 : Enclosure of Active by Well

A_8 : Poly–Poly Spacing

Figure 19.7 Layout of a differential pair with PMOS current-source loads.

19.1.2 Antenna Effect

Suppose the gate of a small MOSFET is tied to a metal 1 interconnect having a large area [Fig. 19.8(a)]. During the etching of metal 1, the metal area acts as an "antenna," collecting ions and rising in potential. It is therefore possible that the gate voltage of the MOS device increases so much that the gate oxide breaks down (irreversibly) during fabrication.

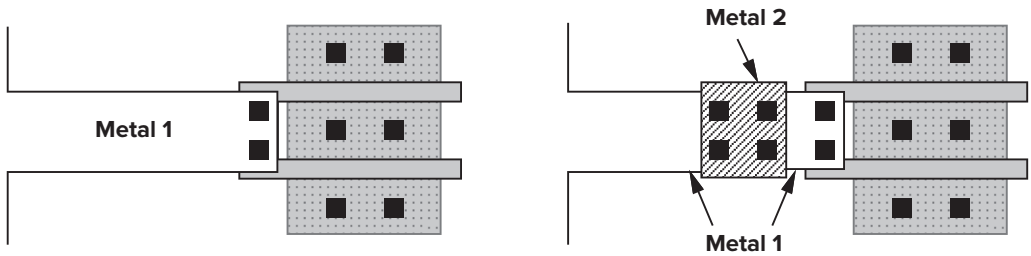


Figure 19.8 (a) Layout susceptible to antenna effect; (b) discontinuity in metal 1 layer to avoid antenna effect.

The antenna effect may occur for any large piece of conductive material tied to the gate, including polysilicon itself. For this reason, submicron CMOS technologies typically limit the total area of such geometries, thereby minimizing the probability of gate-oxide damage. If large areas are inevitable, then a discontinuity can be created as illustrated in Fig. 19.8(b) so that, when metal 1 is being etched, the large area is not connected to the gate.

19.2 ■ Analog Layout Techniques

The extensive sets of design rules enforced by mainstream CMOS processes aim to maximize the yield of digital ICs while allowing moderately aggressive circuit design. Analog systems, on the other hand, demand many more layout precautions so as to minimize effects such as crosstalk, mismatches, noise, etc.

19.2.1 Multifinger Transistors

As mentioned in Chapter 2, wide transistors are usually "folded" so as to reduce both the S/D junction area and the gate resistance. A simple folded structure such as that in Fig. 19.9(a) may prove inadequate for very wide devices, necessitating the use of multiple "fingers" [Fig. 19.9(b)]. As a rule of thumb, the width of each finger is chosen such that the resistance of the finger is less than the inverse transconductance associated with the finger. In low-noise applications, the gate resistance must be one-fifth to one-tenth of $1/g_m$.

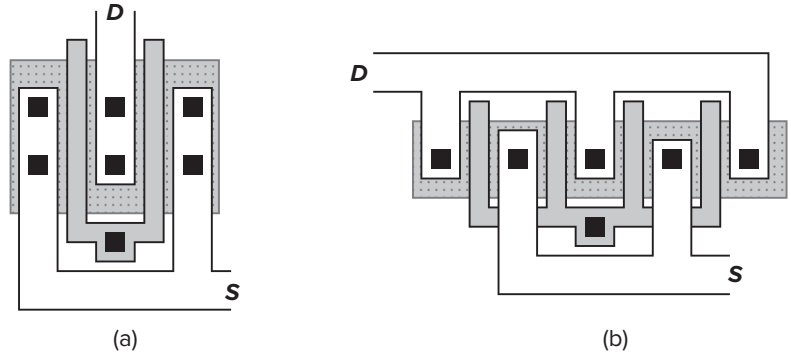


Figure 19.9 (a) Simple folding of a MOSFET; (b) use of multiple fingers.

▶ **Example 19.1**

A 5-μm/40-nm MOSFET biased at 1 mA exhibits a transconductance of $1/(100\ \Omega)$. If the sheet resistance of the gate polysilicon is equal to 30 $\Omega/\square$, what is the widest finger that the structure can incorporate while ensuring that the gate thermal noise voltage is one-fifth of the gate-referred channel thermal noise voltage?

Solution

If the transistor is laid out as N parallel fingers, each finger exhibits a distributed resistance of $30\ \Omega \times (5/0.04)/N$. Using the gate-referred channel thermal noise from Chapter 7, we have for the overall transistor

$$\text{Channel Noise} = \sqrt{4kT\gamma(100)}\ \text{V}/\sqrt{\text{Hz}} \tag{19.1}$$

$$\text{Gate Noise} = \sqrt{4kT\frac{150}{0.04N^2}\frac{1}{3}}\ \text{V}/\sqrt{\text{Hz}} \tag{19.2}$$

where the factor $1/3$ on the right-hand side of (19.2) accounts for the distributed nature of the resistance (Chapter 7). Equating (19.1) to five times (19.2) and assuming that $\gamma = 1$, we have

$$N = 17.7 \tag{19.3}$$

Thus, a minimum of 18 fingers is required. ◀

While the gate resistance can be reduced by decomposing the transistor into more parallel fingers, the capacitance associated with the perimeter of the source/drain areas increases. As exemplified by the structures depicted in Fig. 19.10,[1] with three fingers, the total perimeter of the source or the drain is equal to $2(2E + 2W/3) = 4E + 4W/3$, whereas with five fingers, it is equal to $3(2E + 2W/5) = 6E + 6W/5$.

[1]The use of multiple fingers is sometimes called "interdigitization."

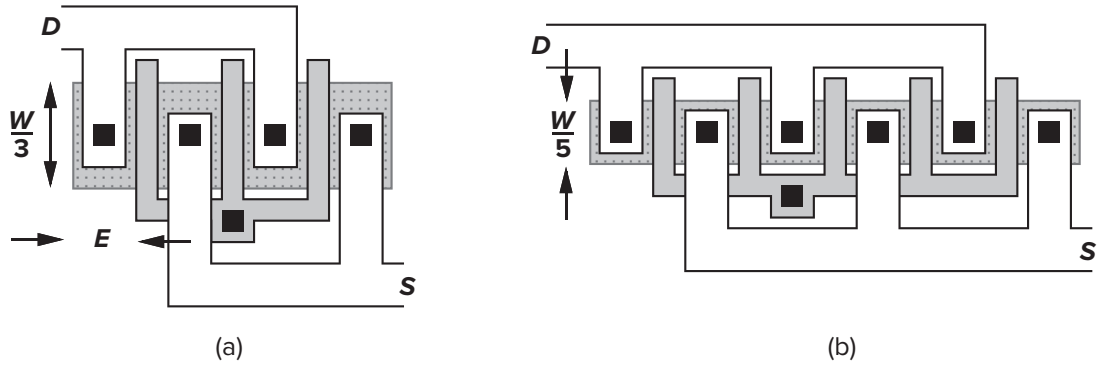


Figure 19.10 Layout of a transistor using (a) three fingers and (b) five fingers.

In general, for an odd number of fingers N, the S/D perimeter capacitance is given by

$$C_P = \frac{N+1}{2}\left(2E + \frac{2W}{N}\right) C_{jsw} \tag{19.4}$$

$$= \left[(N+1)E + \frac{N+1}{N}W\right] C_{jsw} \tag{19.5}$$

Thus, the number of fingers multiplied by E must be much less than W so as to minimize the S/D perimeter capacitance contribution. In practice, this requirement may conflict with that for minimizing the gate resistance noise, demanding a compromise between the two or contacting the gate on both ends to reduce the resistance.

For transistors having a large number of gate fingers, the structure may be modified to that shown in Fig. 19.11, thereby avoiding long geometries and hence disproportionate dimensions in the layout of the overall circuit.

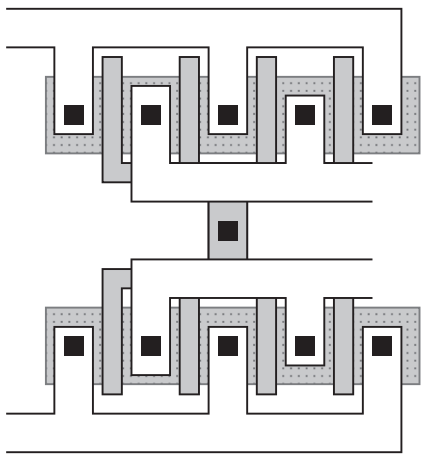


Figure 19.11 Layout of a wide transistor with many fingers.

The layout of a cascode circuit can be simplified if the input device M_1 and the cascode device M_2 have equal widths. As shown in Fig. 19.12(a), the drain of M_1 and the source of M_2 can share the same junction. More important, since this junction is not connected to any other node, it need not accommodate a contact window and can therefore be quite a lot smaller [Fig. 19.12(b)]. Consequently, the capacitance at

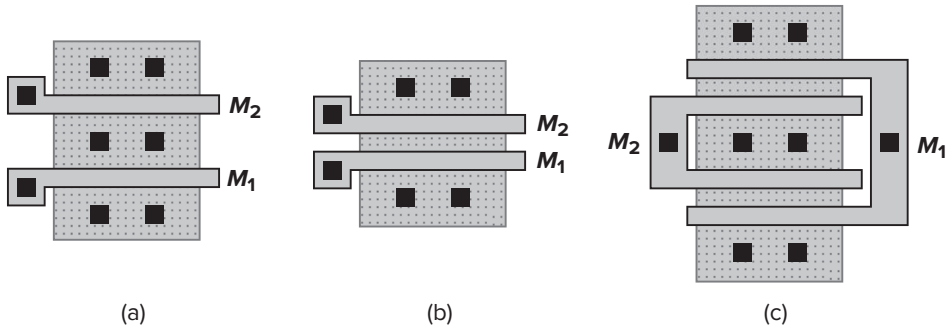


Figure 19.12 Layout of cascode devices having the same width.

the drain of M_1 is reduced substantially, improving the high-frequency performance. For wide transistors, each transistor may use two or more fingers [Fig. 19.12(c)].

19.2.2 Symmetry

Recall from Chapter 14 that asymmetries in fully differential circuits introduce input-referred offsets, thus limiting the minimum signal level that can be detected. While some mismatch is inevitable, inadequate attention to symmetry in the layout may result in large offsets—much greater than the values predicted by the statistical treatment of Chapter 14. Symmetry also suppresses the effect of common-mode noise and even-order nonlinearity. It is important to note that symmetry must be applied to both the devices of interest and their surrounding environment. We return to this point later.

Let us consider the differential pair of Fig. 19.13(a) as the starting point. If, as depicted in Fig. 19.13(b), the two transistors are laid out with different orientations, the matching suffers greatly because many steps in lithography and wafer processing behave differently along different axes.

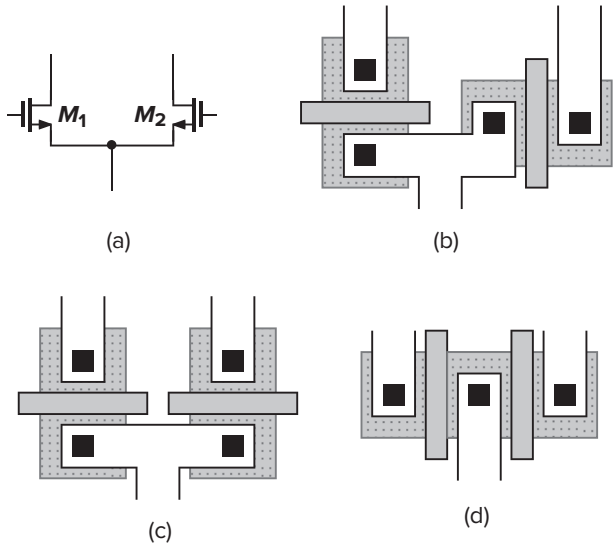


Figure 19.13 (a) Differential pair; (b) layout of M_1 and M_2 with different orientations; (c) layout with gate-aligned devices; (d) layout with parallel-gate devices.

Thus, one of the configurations in Fig. 19.13(c) and (d) provides a more plausible solution. The choice between these two is determined by a subtle effect called "gate shadowing." Illustrated in Fig. 19.14, the

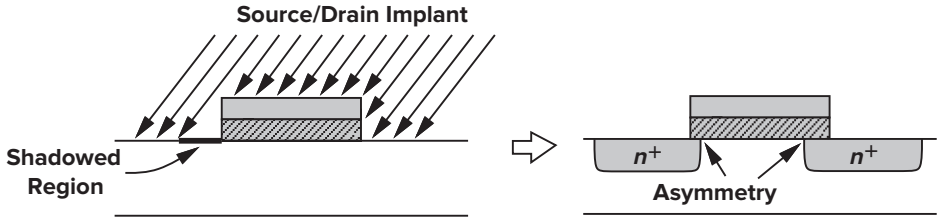


Figure 19.14 Shadowing due to implant tilt.

shadowing is caused by the gate polysilicon during the source/drain implantation because the implant (or the wafer) is tilted by about 7° to avoid channeling (Chapter 18). As a result, a narrow strip in the source or drain region receives less implantation, creating a small asymmetry between the source and drain side diffusions after the implanted areas are annealed.

Now consider the structures of Figs. 19.13(c) and (d) in the presence of gate shadowing (Fig. 19.15). In Fig. 19.15(a), if the shadowed terminal is distinguished as the drain (or the source), then the two devices sustain no asymmetry resulting from shadowing. In Fig. 19.15(b), on the other hand, the transistors are not identical even if the shadowed terminals are distinguished because the source region of M_1 "sees" M_2 to its right, whereas the source region of M_2 sees only the field oxide. Similarly, the drains of M_1 and M_2 see different structures to their left. In other words, the surrounding environment of M_1 is not identical to that of M_2. For this reason, the topology of Fig. 19.15(a) is preferable.

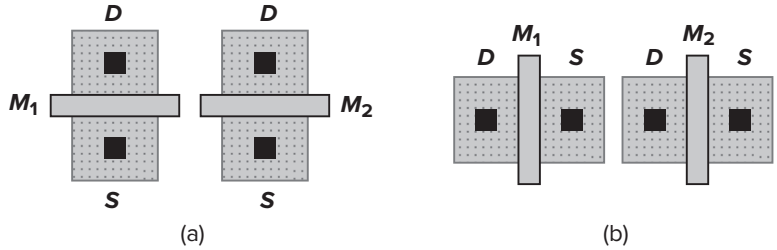


Figure 19.15 Effect of shadowing on (a) gate-aligned and (b) parallel-gate transistors.

The asymmetry inherent in the structures of Fig. 19.15(b) can be ameliorated by adding "dummy" transistors to the two sides so that M_1 and M_2 see approximately the same environment (Fig. 19.16). However, in more complex circuits, e.g., in a folded-cascode op amp, such measures cannot be easily applied. We will see later that a simpler version of dummies proves useful and essential in today's technologies.

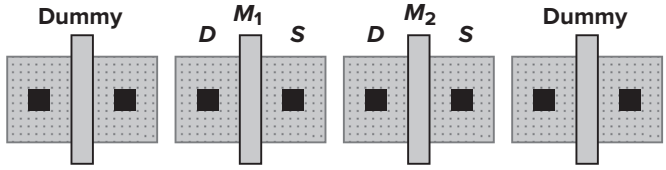


Figure 19.16 Addition of dummy devices to improve symmetry.

We should emphasize the importance of maintaining the same environment on the two sides of the axis of symmetry. For example, in the structure of Fig. 19.17, an unrelated metal line passing over only

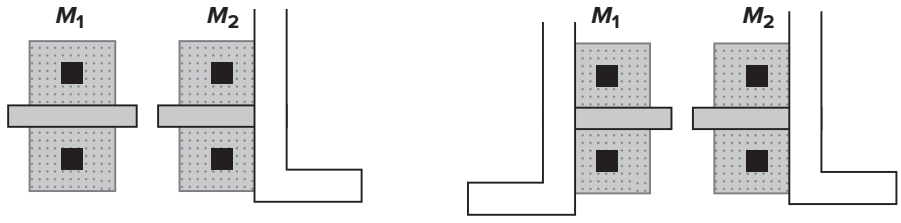


Figure 19.17 (a) Asymmetry resulting from a metal line passing over M_2; (b) removing the asymmetry by replicating the line on top of M_1.

one transistor indeed degrades the symmetry, increasing the mismatch between M_1 and M_2. In such cases, either a replica must be produced on the other side [Fig. 19.17(b)] (even though the replica may be grounded) or, preferably, the source of asymmetry must be removed.

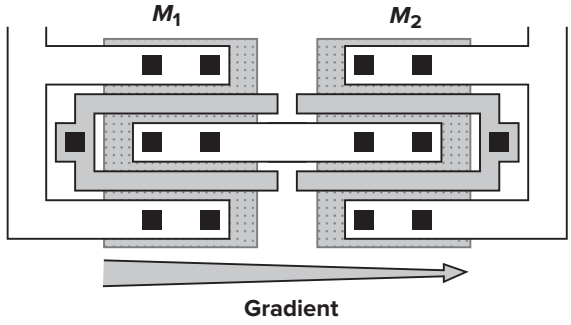


Figure 19.18 Effect of gradient in a differential pair.

Symmetry becomes more difficult to establish for large transistors. In the differential pair of Fig. 19.18, for example, the two transistors have a large width so as to achieve a small input offset voltage, but gradients along the x axis give rise to appreciable mismatches. To reduce the error, a "common-centroid" configuration may be used such that the effect of first-order gradients along both axes is canceled. Illustrated in Fig. 19.19, the idea is to decompose each transistor into two halves that are placed diagonally

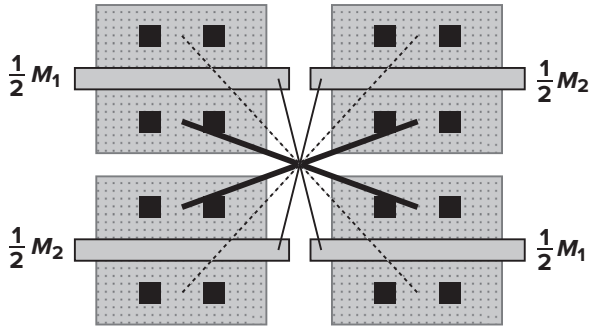


Figure 19.19 Common-centroid layout.

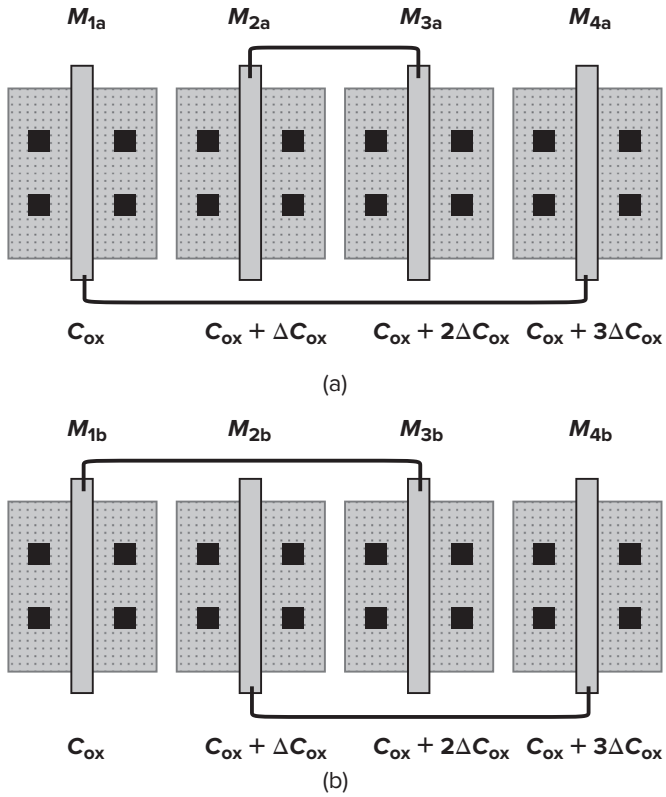


Figure 19.20 One-dimensional cross-coupling.

opposite each other and connected in parallel.[2] However, the routing of interconnects in this layout is quite difficult, often leading to systematic asymmetries of the type depicted in Fig. 19.17(a) or in the capacitances from the wires to ground and between the wires. For a larger circuit, e.g., an op amp, the routing may become prohibitively complex. We thus seek simpler solutions.

The effect of linear gradients can be suppressed by "one-dimensional" cross-coupling, as depicted in Fig. 19.20. Here, all four half transistors are placed along the same axis, and M_1 and M_2 are formed by connecting either the near ones and the far ones [Fig. 19.20(a)] or every other one [Fig. 19.20(b)]. (For clarity, the connections between the sources and the drains are not shown.) To analyze the effect of gradients in these structures, let us assume that, for example, the gate-oxide capacitance varies by ΔC_{ox} from each half transistor to the next.[3] Placing M_{1a} and M_{4a} in parallel, we have

$$I_{D1a} + I_{D4a} = \frac{1}{2}\mu_n(C_{ox} + C_{ox} + 3\Delta C_{ox})\frac{W}{L}(V_{GS} - V_{TH})^2 \tag{19.6}$$

and for M_{2a} and M_{3a},

$$I_{D2a} + I_{D3a} = \frac{1}{2}\mu_n(C_{ox} + \Delta C_{ox} + C_{ox} + 2\Delta C_{ox})\frac{W}{L}(V_{GS} - V_{TH})^2 \tag{19.7}$$

[2]The interconnect lines shown in this figure are only conceptually correct.

[3]In reality, variation of C_{ox} influences the threshold voltage as well. We neglect this effect here.

This type of cross-coupling therefore cancels the effect of the gradient. Now, for the configuration of Fig. 19.20(b), we have

$$I_{D1b} + I_{D3b} = \frac{1}{2}\mu_n(C_{ox} + C_{ox} + 2\Delta C_{ox})\frac{W}{L}(V_{GS} - V_{TH})^2 \tag{19.8}$$

and

$$I_{D2b} + I_{D4b} = \frac{1}{2}\mu_n(C_{ox} + \Delta C_{ox} + C_{ox} + 3\Delta C_{ox})\frac{W}{L}(V_{GS} - V_{TH})^2 \tag{19.9}$$

Equations (19.8) and (19.9) suggest that this approach removes the error to a lesser extent.

The reader can prove that for small gradients in other device parameters, similar results are obtained, concluding that the topology of Fig. 19.20(a) contains smaller errors than that of Fig. 19.20(b). However, since the environment seen by $M_{2a} + M_{3a}$ differs from that seen by $M_{1a} + M_{4a}$, dummy transistors must be added to the left of M_{1a} and the right of M_{4a}.

19.2.3 Shallow Trench Isolation Issues

Modern MOS devices are surrounded by a shallow "trench" so as to avoid the formation of a channel between adjacent transistors [Fig. 19.21(a)]. Called "shallow trench isolation" (STI) and created automatically, this structure is filled with oxide and exhibits a different thermal expansion coefficient from that of silicon. As a result, during fabrication steps, the STI and the enclosed silicon area expand and contract differently. This STI-induced "stress" alters the electrical properties of the MOS transistor, introducing substantial error in its I/V characteristics.

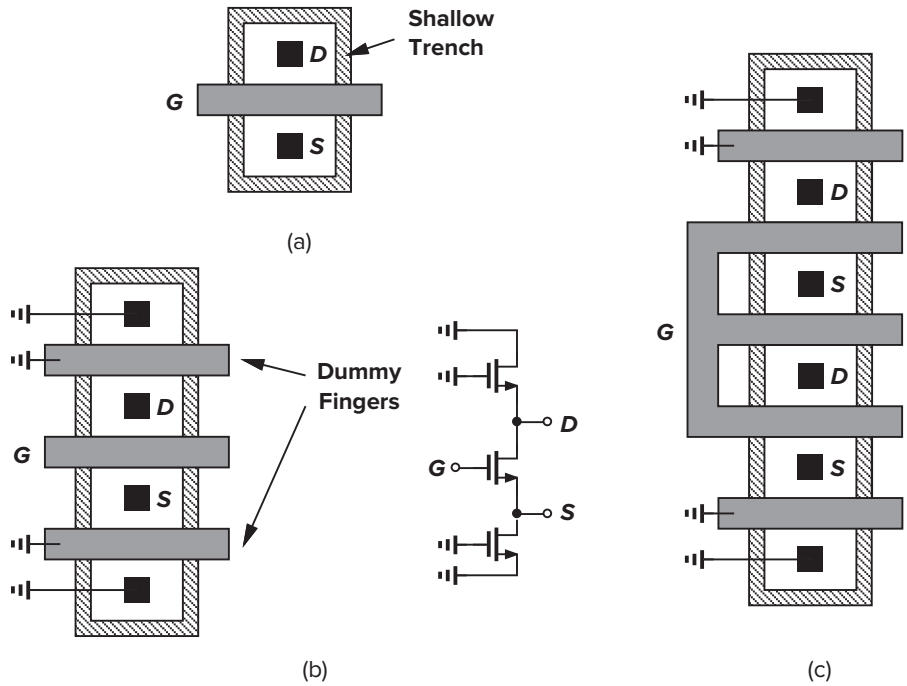


Figure 19.21 (a) Shallow trench isolation surrounding a device, (b) use of dummy fingers to reduce STI-induced stress, and (c) multifinger transistor example.

In order to alleviate this issue, we must minimize the propagation of the stress toward the gate area. To this end, we insert two fingers on the two sides of the main device [Fig. 19.21(b)]. These "dummy" fingers and their associated S/D junctions are typically grounded to ensure that they do not interfere with the operation of the main transistor. Note, however, that the dummy gates increase the S/D capacitances to ground.

For transistors employing multiple fingers, the dummy gates can be simply added to the two ends of the array [Fig. 19.21(c)].

19.2.4 Well Proximity Effects

As explained in Chapter 18, an n-well is formed by an N-type implant onto the exposed areas of silicon. The unexposed areas are covered by a thick layer consisting of oxide and photoresist [Fig. 19.22(a)]. Unfortunately, the implant does not occur at a 90° angle with respect to the wafer, thus reflecting from the walls formed by oxide and photoresist and creating *nonuniform* doping in the n-well. That is, the border areas of the n-well receive a different doping density from those in the middle of the n-well. Consequently, the PMOS devices located near the *edges* of the n-well have different I/V characteristics compared to those in the middle. We call this effect the "well proximity" error. For example, the current mirror arrangement shown in Fig. 19.22(b) exhibits mismatches between M_1 and M_2 or M_3 because M_1 is more heavily influenced by the implant reflections.

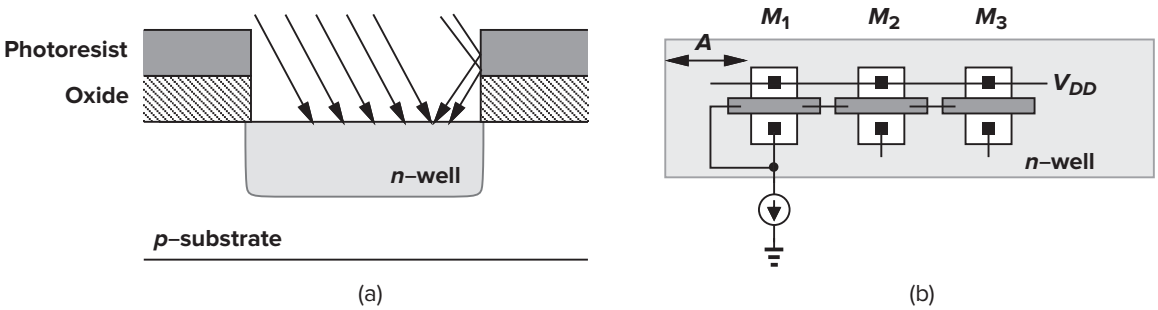


Figure 19.22 (a) Effect of implant reflection on n-well doping uniformity, and (b) current mirror arrangement showing the effect of n-well edges.

To reduce the well proximity effect, the n-well must extend well beyond the PMOS devices. For example, A in Fig. 19.22(b) can be chosen greater than several microns.

19.2.5 Reference Distribution

In analog systems, the bias currents and voltages of various building blocks are derived from one or more bandgap reference generators. The distribution of such references across a large chip entails a number of important issues. Consider the example depicted in Fig. 19.23, where I_{REF} is produced by a bandgap reference and M_1–M_n serve as bias current sources of building blocks that are located far from M_{REF} and from each other. If the matching between I_{D1}–I_{Dn} and I_{REF} is critical, then the voltage drop along the ground line must be taken into account. In fact, for a large number of circuits connected to the same ground line, the systematic mismatch between the current sources and I_{REF} may be unacceptable.

To remedy the above difficulty, the reference can be distributed in the current domain rather than in the voltage domain. Illustrated in Fig. 19.24, the idea is to route the reference current to the vicinity of the

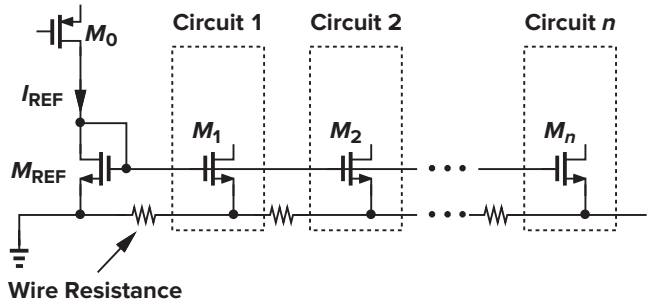


Figure 19.23 Distribution of a reference voltage for current mirror biasing.

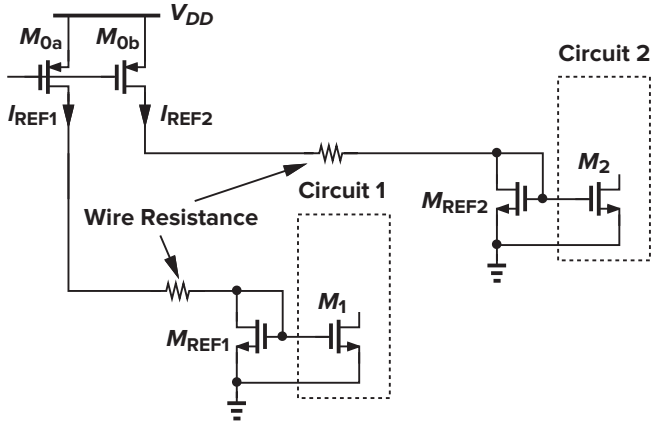


Figure 19.24 Distribution of current to reduce the effect of interconnect resistance.

building blocks and perform the current mirror operation *locally*. At the destination, bypass capacitors suppress any noise that the long interconnects may pick up. Placing the interconnect resistance in series with current sources, this approach lowers systematic errors if the building blocks appear in dense groups in different regions on the chip. However, mismatches between I_{REF1} and I_{REF2} and between M_{REF1} and M_{REF2} introduce error. In large systems, it may be advantageous to employ several local bandgap reference circuits so as to alleviate routing problems.

Another issue in the circuits of Figs. 19.23 and 19.24 relates to the orientation of the transistors. As mentioned in Sec. 19.2.2, if, for example, M_{REF} and M_1–M_n in Fig. 19.23 have different orientations, then substantial mismatches arise. Since circuits $1, 2, \ldots, n$ may be laid out individually, particular attention must be paid to the orientation of their current sources while the entire chip is assembled.

The scaling of currents in Figs. 19.23 and 19.24 also demands careful choice of device dimensions and layout. Suppose the circuit of Fig. 19.23 requires $I_{D1} = 0.5I_{REF}$ and $I_{D2} = 2I_{REF}$. How do we choose $(W/L)_1$ and $(W/L)_2$ with respect to $(W/L)_{REF}$? Recall from Chapter 2 that, owing to the side diffusion of the source/drain regions, the effective channel length is less than the drawn length by $2L_D$, a poorly controlled quantity. Thus, to avoid large mismatches, the lengths of the transistors must be equal and the currents must be scaled by proper choice of the widths. We then choose that $W_1 = 0.5W_{REF}$ and $W_2 = 2W_{REF}$. Figure 19.25 shows how M_{REF}, M_1, and M_2 in this example are laid out to ensure reasonable matching. Note that all equivalent widths are integer multiples of a unit value, W_u. Transistor M_1 is identical to M_{REF} except that half of its source remains floating (or connected to the drain). To improve the matching, the array can be surrounded by dummy devices.

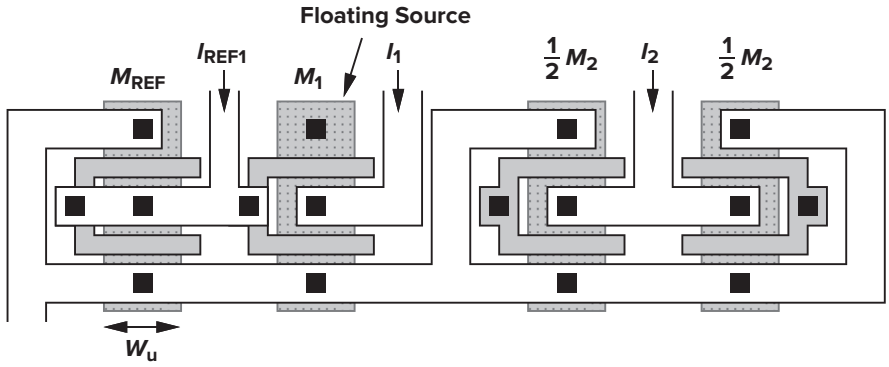


Figure 19.25 Proper scaling of device dimensions for adequate matching of current sources.

19.2.6 Passive Devices

Resistors Polysilicon resistors using a silicide block exhibit high linearity, low capacitance to the substrate, and relatively small mismatches. The linearity of these resistors in fact depends on their length [1], necessitating accurate measurement and modeling for high-precision applications. Figure 19.26 depicts an example in which the nonlinearity of the resistor is critical. Since $V_{out} = -I_{in}R_F$, the accuracy of current-to-voltage conversion depends on the linearity of R_F. In practice, however, the op amp limits the linearity.

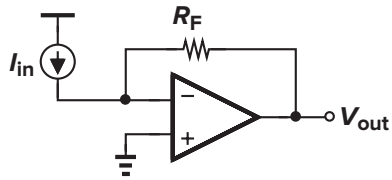


Figure 19.26 Feedback amplifier converting a voltage to current.

As with other devices, the matching of polysilicon resistors is a function of their dimensions. For example, resistors having a length and width of a few microns display typical mismatches on the order of 0.2%. Most of the symmetry rules described for the layout of MOS devices apply to resistors as well. For example, resistors that are required to bear a well-defined ratio must consist of identical units placed in parallel or in series (with the same orientation).

▶ **Example 19.2**

Consider the bandgap circuit shown in Fig. 19.27. Choose the values of n, R_1, and R_2 such that V_{out} exhibits a zero temperature coefficient and the layout can be designed for high precision.

Solution

Since $V_{out} = V_{BE3} + V_T(R_2/R_1)\ln n$, we must find convenient values of n, R_1, and R_2 such that $(R_2/R_1)\ln n \approx 17.2$ (Chapter 12). If $n = 31$, then $R_2/R_1 \approx 5$, yielding the layout of Fig. 19.28(a). Note that R_1 is placed in the middle to partially cancel the effect of gradients.

Now suppose we choose $n = 25$, obtaining $R_2/R_1 = 5.34$. Such a value cannot be accurately established by simply adjusting the dimensions of R_2 and R_1. Rather, we write $R_2/R_1 = 16/3$ and construct the resistors as shown in Fig. 19.28(b).

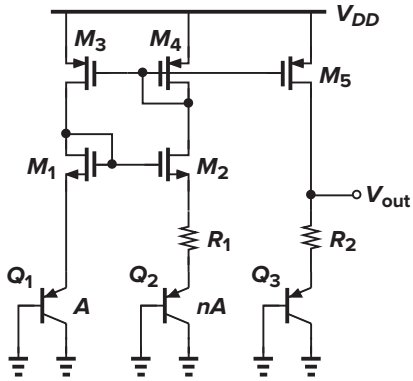


Figure 19.27

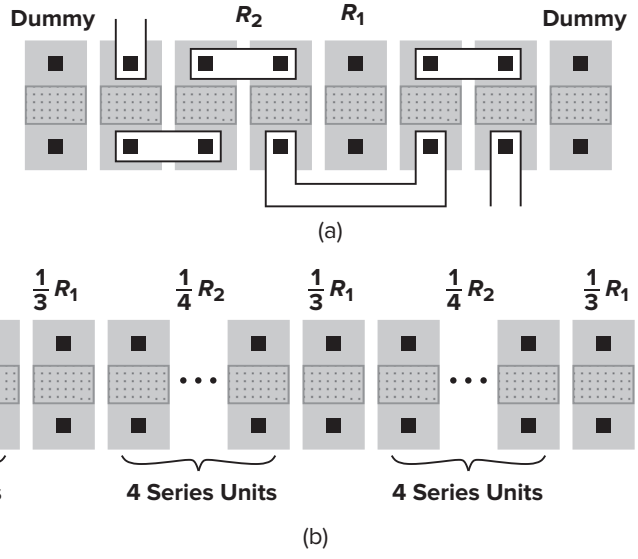


Figure 19.28 Layout of R_1 and R_2 with (a) $R_2/R_1 = 5$ and (b) $R_2/R_1 = 5.34 \approx 16/3$.

The resistance of the polysilicon structure studied above consists of two components: that due to the unsilicided region and the resistance associated with the two contacts. As depicted in Fig. 19.29(a), the

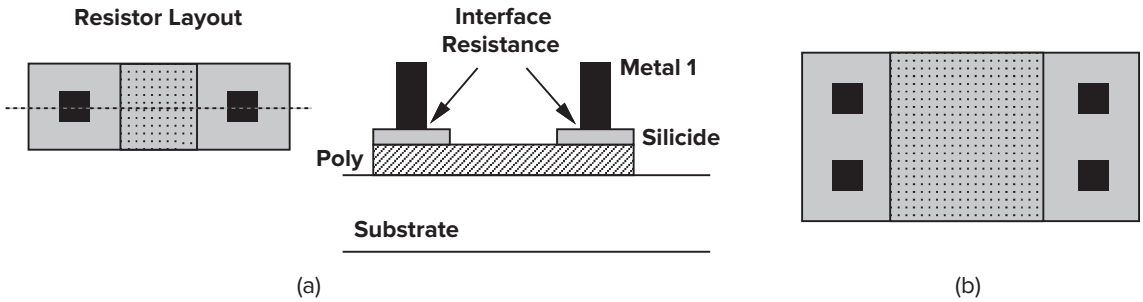


Figure 19.29 (a) Top view and cross section of a poly resistor, and (b) doubling the width and length to reduce the effect of contact resistance.

narrow contact window (about 80 nm×80 nm in 40-nm technology) results in a high interface resistance between metal 1 and the silicide area. This component is poorly controlled and must preferably remain much less than the first. For example, the length and width of the structure in Fig. 19.29(a) can be doubled so as to halve the total contact resistance while keeping the unsilicided region's resistance approximately constant [Fig. 19.29(b)].

For large values, resistors are usually decomposed into shorter units that are laid out in parallel and connected in series [Fig. 19.30(a)]. From the viewpoint of matching and reproducibility, this structure is preferable to "serpentine" topologies [Fig. 19.30(b)], where the corners contribute significant resistance.

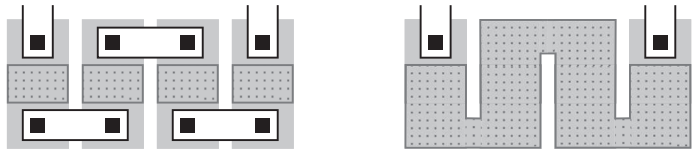


Figure 19.30 (a) Layout of large resistors; (b) serpentine topology.

The sheet resistance, $R_\square$, of polysilicon resistors varies with temperature and process, necessitating provisions in the design for this variation. The temperature coefficient depends on the doping type and level and must be measured for each technology. Typical values are $+0.1\%$ /°C and -0.1% /°C for p^+ and n^+ doping, respectively. The variation with process is usually less than $\pm 20\%$.

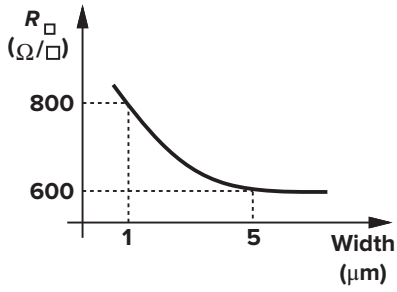


Figure 19.31 Dependence of n-well sheet resistance upon resistor width.

In technologies lacking a silicide block mask, resistors may be made of n-well, source/drain p^+ or n^+ material, silicided polysilicon, or metal, with $R_\square$ decreasing in this order. The sheet resistance of n-well is typically around 1 kΩ, but it may vary by a large fraction, e.g., $\pm 40\%$, with process. Furthermore, $R_\square$ depends on the *width* of the resistor, as exemplified by the plot of Fig. 19.31. This is because, with a depth of several microns, n-well regions exhibit width-dependent diffusion at the edges. Also, $R_\square$ is a strong function of the n-well–substrate voltage difference, giving rise to both nonlinearity and poor definition of the value of the resistor. For example, in the circuit of Fig. 19.32, resistors R_S and R_D suffer from large mismatches in $R_\square$ because the depletion region below R_S is quite a lot narrower than that below R_D. Also, as V_{out} varies, so does the sheet resistance of R_D, introducing nonlinearity. Resistors made of n-well display a TC of $+0.2\%$ to $+0.5\%$ /°C.

▶ Example 19.3

An A/D converter incorporates a resistor ladder consisting of 128 units made of n-well to generate equally-spaced reference voltages (Fig. 19.33). If the two ends of the ladder are connected to $V_1 = +1$ V and $V_2 = +2$ V, calculate the ratio R_{128}/R_1.

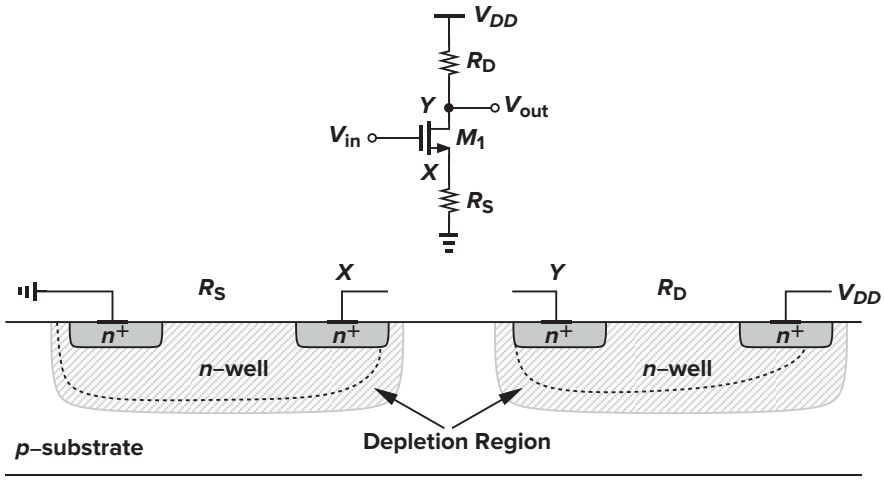


Figure 19.32 Common-source stage using n-well resistors.

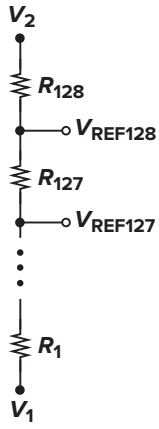


Figure 19.33 Resistor ladder used in an A/D converter.

Solution

The width of the depletion region inside the n-well is given by $x_d = \sqrt{2\epsilon_{si}(\phi_B + V_R)/(qN_{well})}$, where N_{well} denotes the n-well doping level and V_R the reverse bias voltage. Assuming that the zero-bias depth of the n-well is equal to t_0, we have

$$\frac{R_{128}}{R_1} = \frac{t_0 - \sqrt{\frac{2\epsilon_{si}}{qN_{well}}(\phi_B + V_1)} + \sqrt{\frac{2\epsilon_{si}}{qN_{well}}\phi_B}}{t_0 - \sqrt{\frac{2\epsilon_{si}}{qN_{well}}(\phi_B + V_2)} + \sqrt{\frac{2\epsilon_{si}}{qN_{well}}\phi_B}} \tag{19.10}$$

$$= \frac{t_0 + \sqrt{\frac{2\epsilon_{si}}{qN_{well}}\phi_B}\left(1 - \sqrt{1 + \frac{V_1}{\phi_B}}\right)}{t_0 + \sqrt{\frac{2\epsilon_{si}}{qN_{well}}\phi_B}\left(1 - \sqrt{1 + \frac{V_2}{\phi_B}}\right)} \tag{19.11}$$

If the difference between R_1 and R_{128} is small, we can divide the numerator and denominator of (19.11) by t_0 and approximate the result as

$$\frac{R_{128}}{R_1} \approx \left[1 + \frac{1}{t_0}\sqrt{\frac{2\epsilon_{si}}{qN_{well}}\phi_B}\left(1 - \sqrt{1 + \frac{V_1}{\phi_B}}\right)\right]\left[1 - \frac{1}{t_0}\sqrt{\frac{2\epsilon_{si}}{qN_{well}}\phi_B}\left(1 - \sqrt{1 + \frac{V_2}{\phi_B}}\right)\right] \tag{19.12}$$

$$\approx 1 + \frac{1}{t_0}\sqrt{\frac{2\epsilon_{si}}{qN_{well}}\phi_B}\left(\sqrt{1 + \frac{V_2}{\phi_B}} - \sqrt{1 + \frac{V_1}{\phi_B}}\right) \tag{19.13}$$

For example, if $t_0 = 2\ \mu$m, $N_{well} = 10^{16}$ cm^{-1}, and $\phi_B = 0.7$ V, the mismatch between R_{128} and R_1 is nearly 60%. ◀

The p^+ and n^+ source/drain regions can also be used as resistors. With a sheet resistance of 20 to 30 ohms per square, silicided S/D regions are suited only to low-value resistors. Furthermore, the junction between these areas and the bulk introduces capacitance and voltage dependence.[4]

Silicided polysilicon has a sheet resistance of 20 to 30 ohms per square and can be utilized for low resistor values. While suffering from less capacitance to the substrate than n^+ or p^+ resistors, silicided polysilicon has a process-dependent $R_\square$, with variations as high as 20 to 30%. Thus, it can be used only if its absolute value is not critical, for example, in the resistor ladder of Fig. 19.33. The temperature coefficient of this type of resistor is between +0.2 and +0.4%/°C.

The metal layers in a process can provide very low resistor values. For example, in extremely high-speed A/D converters, the ladder of Fig. 19.33 may be constructed as simply a long metal line having equally spaced taps (Fig. 19.34). Note, however, that if the width of the metal resistor is small, matching suffers. The temperature coefficient of the resistance is about 0.3%/°C for aluminum.

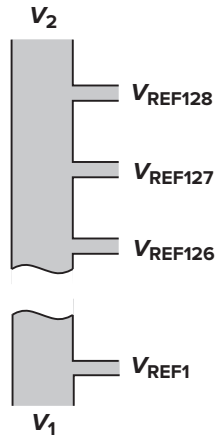


Figure 19.34 Resistor ladder made of metal.

Capacitors As explained in Chapter 18, linear capacitors are designed using sandwiches made of the available conductive layers. For example, in a process having nine layers of metal, the capacitors can be formed as shown in Fig. 19.35. The choice of one topology over another is determined by two factors: (1) the area occupied by the capacitor and (2) the ratio of the bottom-plate parasitic capacitance to the

[4]The nonlinearity of n-well resistors is much higher because the low doping level in the n-well results in a greater sensitivity to the voltage with respect to the substrate.

interplate capacitance, C_P/C. In typical technologies, the capacitance between consecutive metal layers (e.g., C_1 or C_2) in Fig. 19.35(d) is on the order of 35 to 40 aF/μm^2, and that between metal 1 and polysilicon is about 60 aF/μm^2. Thus, the structure of Fig. 19.35(d) provides more than nine times the density of that in Fig. 19.35(a). On the other hand, the value of C_P increases from Fig. 19.35(a) to Fig. 19.35(d). With typical values, C_P/C reaches a minimum—about 5 to 10%—for the structure of Fig. 19.35(b) or (c) and increases to about 20% for the sandwich of Fig. 19.35(d).

Since the absolute value of interlayer capacitances is poorly controlled in digital technologies, the capacitors of Fig. 19.35 may experience process variations as high as 20%. By contrast, the gate-oxide capacitance is typically controlled with less than 5% error. Interestingly, the structure of Fig. 19.35(d) may suffer from less variation than the others because random variations in the capacitances between various layers tend to "average out."

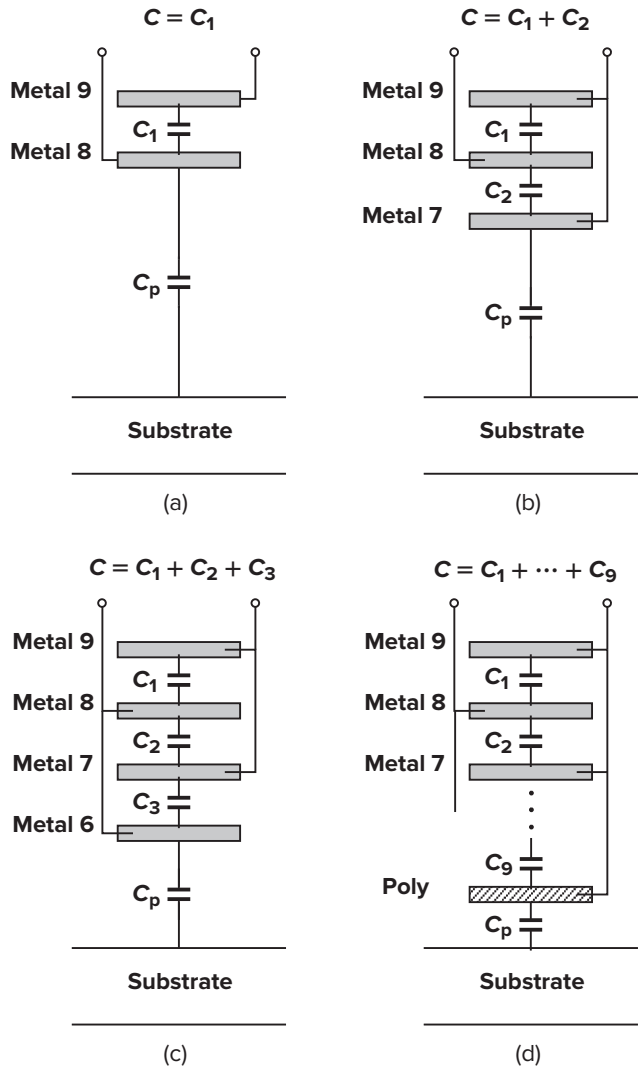


Figure 19.35 Capacitor structures using various conductive layers.

We have thus far neglected the fringe capacitance. As depicted in Fig. 19.36, the electric field lines emanating from the edge of each plate must terminate on the edge of the other plate or on the substrate,

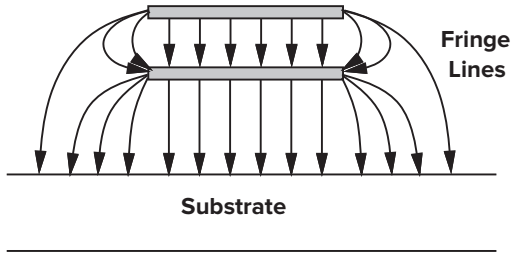


Figure 19.36 Fringe component of capacitance.

giving rise to a fringe capacitance that must be taken into account. The fringe capacitance can be calculated using Eq. (18.11) or from tabulated values in the process design manual.

As explained in Chapter 18, a MOS transistor with its source and drain tied together can act as a capacitor if the gate-source potential is sufficient to establish an inversion layer. However, the voltage dependence of the capacitance limits the use of this structure.

The layout of capacitors for high-precision circuits must follow the principles described earlier for transistors and resistors. For example, in applications where an array of well-matched capacitors is required, dummy devices must be placed on the perimeter of the array.

▶ Example 19.4

The circuit of Fig. 19.37(a) is designed for a nominal gain of $C_1/C_2 = 8$. How should C_1 and C_2 be laid out to ensure precise definition of the gain?

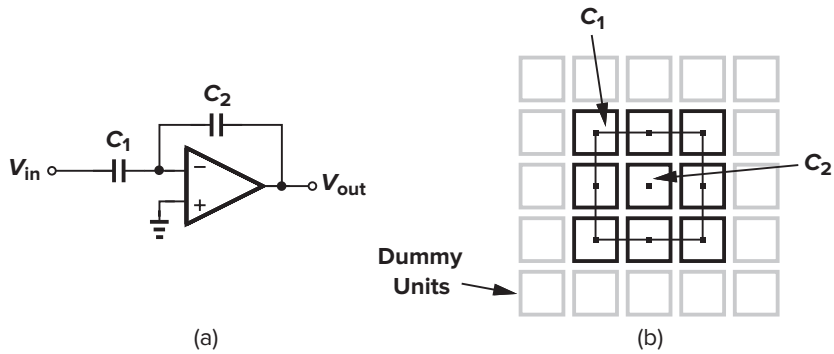


Figure 19.37

Solution

We form C_1 as 8 unit capacitors, each equal to C_2, and place all of the units in a square array [Fig. 19.37(b)]. Note that (1) C_2 is symmetrically surrounded by the units comprising C_1 so that the effect of vertical or horizontal gradients is canceled to the first order; and (2) dummy capacitor units are placed around the main array, creating approximately the same environment for the units of C_1 as that seen by C_2.

◀

For large capacitor arrays, cross-coupling techniques such as those illustrated in Figs. 19.20 and 19.27 can be applied. However, unlike transistors and resistors, capacitors are quite sensitive to the wiring capacitance, demanding great care in the interconnection of the units. Even in the simple array of Fig. 19.37(b), it is difficult to route all of the top-plate and bottom-plate connections while introducing no additional capacitance. As the layout of Fig. 19.38 exemplifies, the wiring inevitably leads to some error in the ratio C_1/C_2.

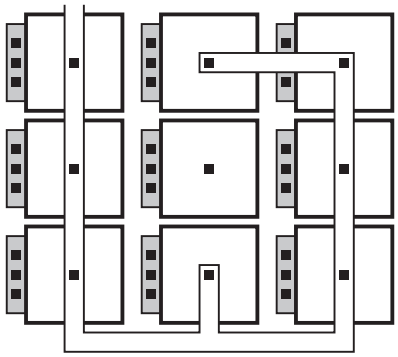


Figure 19.38 Layout of capacitors along with interconnections.

Diodes Two types of *pn* junctions can be formed in a standard CMOS technology: one in the *p*-substrate and another in an *n*-well (Fig. 19.39). The former must remain reverse biased and can therefore serve only as a voltage-dependent capacitor (varactor), e.g., in voltage-controlled oscillators.

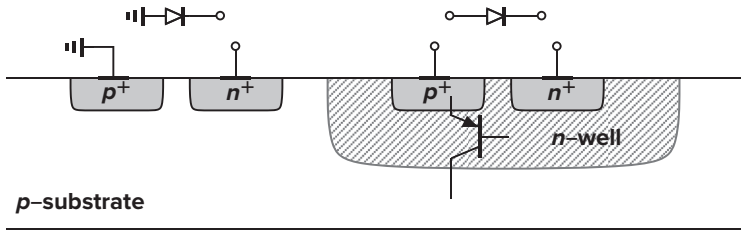


Figure 19.39 Diodes in CMOS technology.

The diode formed in an *n*-well also faces difficulties if forward biased. Recall from Chapter 12 that the p^+ region in the *n*-well, the *n*-well itself, and the *p*-substrate constitute a bipolar *pnp* transistor whose collector is typically grounded. Thus, if the *pn* junction in the *n*-well is forward biased, substantial current flows from the p^+ terminal to the substrate. In other words, the structure must not be viewed as merely a two-terminal floating diode. Nonetheless, if reverse-biased, the device can serve as a varactor.

Owing to these difficulties, analog CMOS circuits rarely incorporate forward-biased diodes (except in bandgap circuits).

19.2.7 Interconnects

Modern CMOS processes offer a dozen metal layers for interconnection, but the cost may dictate the use of eight or nine. Many effects related to wires must be taken into account when a high-precision and/or high-speed circuit is laid out.

Capacitance The parallel-plate and fringe capacitance of wires may degrade the speed if long interconnects are required. For example, in a mixed-signal system (e.g., using many switched-capacitor circuits), the clock signal must be distributed over long wires to access various building blocks, thereby experiencing significant line capacitance. More important, the capacitance between lines introduces substantial coupling of signals.

Figure 19.40 illustrates an example of cross talk between signals. Here, a common-source stage and a NAND gate are located next to each other, and the two inputs to the gate, V_A and V_B, cross over the analog signal, V_{in}. Furthermore, the clock wire, CK, is laid out in parallel with V_{in}, and the output of the NAND gate has some overlap with the output of the common-source stage. Each of the coupling capacitances in this layout may corrupt V_{in} or V_{out}. Note that, even though the coupling capacitances are

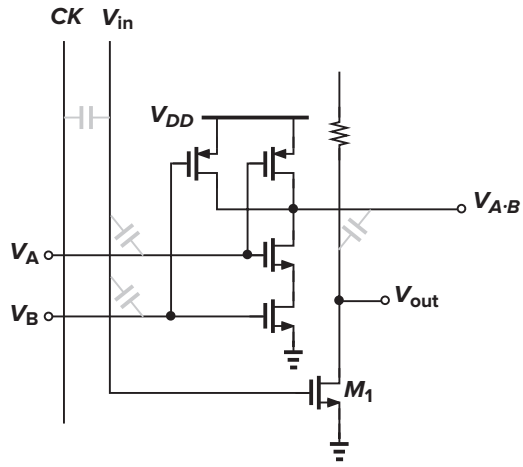


Figure 19.40 Capacitive coupling between various lines in a typical layout.

small, the signal corruption may be appreciable because the voltage swings on V_A, V_B, $V_{A\cdot B}$, and CK are large. For example, if the capacitance between CK and V_{in} is 50 aF, and the total capacitance seen from V_{in} to ground 10 fF, then a 1-V change in CK corrupts V_{in} by 5 mV.

Crosstalk can be reduced through the use of two techniques. First, differential signaling converts most of the crosstalk to common-mode disturbance. For example, if the circuit of Fig. 19.40 is modified to that shown in Fig. 19.41, the coupling of V_A and V_B to V_{in}^+ and V_{in}^- produces no differential error if $C_1 = C_1'$ and $C_2 = C_2'$. Even for 10% mismatch between the capacitances, the differential corruption is one order of magnitude less than that in Fig. 19.40. Note that a dummy wire is added to the layout so as to create an overlap capacitance between CK and V_{in}^- equal to that between CK and V_{in}^+. As mentioned in Chapter 4, it is desirable to employ differential clocks as well to suppress the net coupling further.

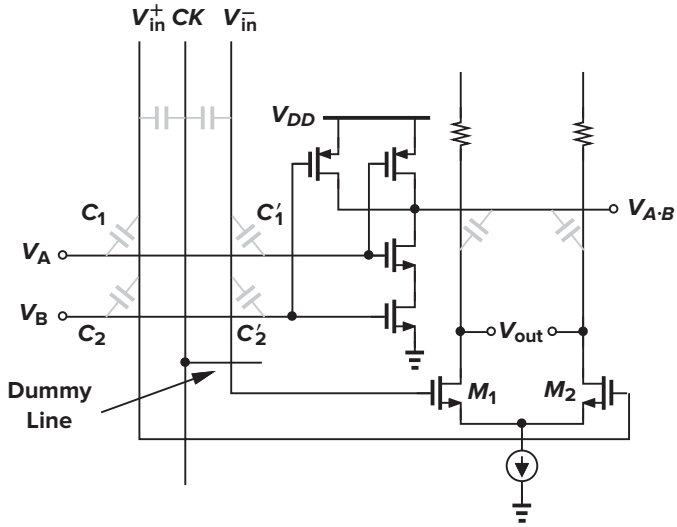


Figure 19.41 Reduction of capacitive coupling through the use of differential signaling.

Second, sensitive signals can be "shielded" in the layout. Depicted in Fig. 19.42(a), one approach places ground lines on the two sides of the signal, forcing most of the electric field lines emanating from the "noisy" lines to terminate on ground rather than on the signal. Note that this method proves more

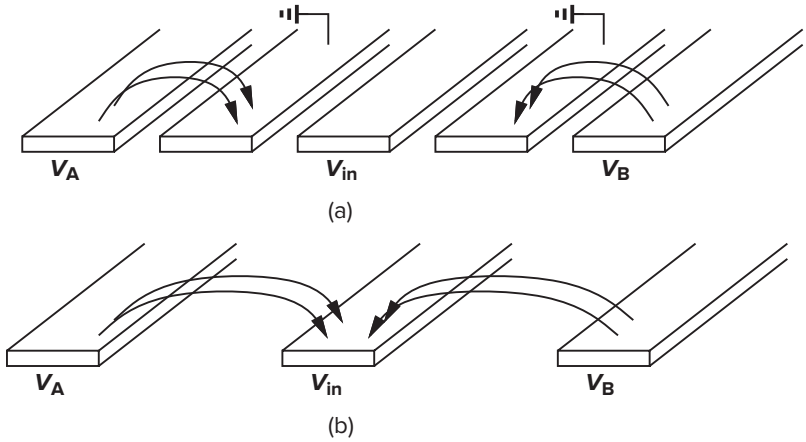


Figure 19.42 (a) Shielding sensitive signals by additional ground lines; (b) greater spacing between lines to reduce coupling.

effective than simply allowing more space between the signal and the noisy lines [Fig. 19.42(b)]. The shielding, however, is obtained at the cost of more complex wiring and greater capacitance between the signals and ground.

Another shielding technique is shown in Fig. 19.43. Here, the sensitive line in metal 6 is surrounded by a grounded shield consisting of a higher and a lower metal layer and hence fully isolated from external electric field lines.[5] However, the signal experiences higher capacitance to ground, and the use of three metal layers here complicates the routing of other signals.

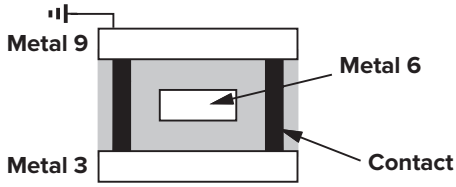


Figure 19.43 Shielding a sensitive line (metal 2) by lower and upper ground planes.

Resistance The resistance of interconnects also requires attention. In low-noise applications, long signal wires—with sheet resistances of 40 to 80 m$\Omega/\square$—may introduce substantial thermal noise. Furthermore, the contacts and vias also suffer from a high resistance. For example, an 80-nm $\times$ 80-nm metal contact to silicided polysilicon exhibits a resistance of 30 to 40 Ω, and a via between metal 1 and metal 2, a resistance of 5 to 10 Ω.

▶ Example 19.5

In the layout of Fig. 19.44, a 100-μm metal 4 line is connected to a sequence of vias and contacts to reach the gate of a transistor. Calculate the thermal noise contributed by the line and the contacts.

Solution

Assuming $R_\square = 40$ m$\Omega/\square$ for metal 4, a via resistance of 5 Ω, and a poly contact resistance of 30 Ω, we have $R_{tot} = 2 + 2.5 + 2.5 + 2.5 + 15 = 24.5\ \Omega$. The thermal noise voltage is thus equal to 0.64 nV/$\sqrt{\text{Hz}}$ at room

[5] We assume that the ground connection itself does not contain noise. We return to this issue in Sec. 19.4.

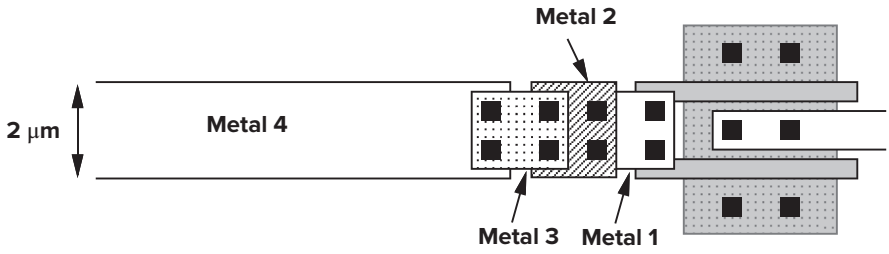


Figure 19.44

temperature. If guiding the input signal to a low-noise amplifier, this interconnect arrangement considerably raises the input-referred noise.

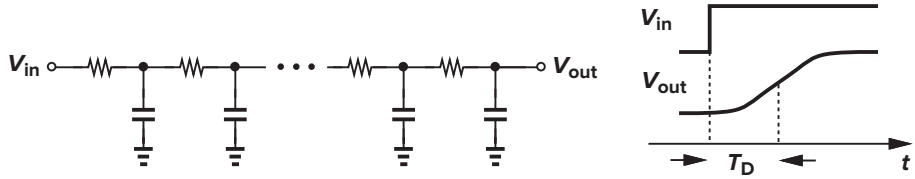


Figure 19.45 Delay and dispersion of a signal in a long line.

The distributed resistance and capacitance of long interconnects may introduce significant delay and "dispersion" in signals. Illustrated in Fig. 19.45, the delay can be approximated as

$$T_D = \frac{1}{2} R_u C_u L^2 \tag{19.14}$$

where R_u and C_u denote the resistance and capacitance per unit length, respectively, and L is the total length. For example, consider the circuit shown in Fig. 19.46, where an array of samplers senses the analog input V_{in} and is activated by CK. If the delays experienced by CK and V_{in} from the left side to the right side are unequal, then so are the levels sampled by $C_1, \ldots, C_n$, distorting the sampled waveform. Even if the clock and signal lines and their capacitive loading are identical, CK and V_{in} may still suffer from unequal delays because the former is a rectangular wave and the latter is not.

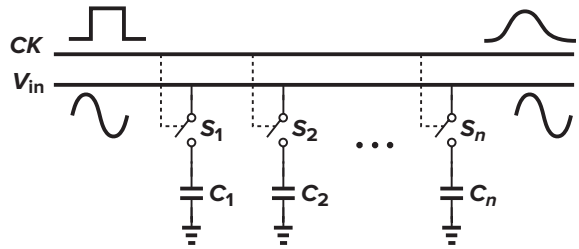


Figure 19.46 An array of sampling circuits sensing an input.

The term "dispersion" refers to the significant increase in the transition time of the signal as it propagates through a line, a particularly troublesome effect if a clock edge is to define a sampling point. In the example of Fig. 19.46, the clock waveform applied to S_n displays long rise and fall times, making the sampling susceptible to both noise and distortion [4]. The clock edges can be sharpened by inserting an inverter

between CK and every switch, but at the cost of greater uncertainty in the delay difference between CK and V_{in}.

As mentioned in Chapter 18, the design of power and ground buses on a chip requires attention to a number of issues. In large ICs, the dc or transient voltage drop along the buses may be significant, affecting sensitive circuits supplied by the same lines. Furthermore, electromigration calls for a minimum line width to guarantee long-term reliability. With the multiple interconnect levels available in today's CMOS technology, it is possible to connect two or more layers in parallel, thereby reducing the series resistance and alleviating electromigration constraints. Since the thickness of the top metal layer is typically twice that of the lower ones, at least *three* layers must be placed in parallel to relax these issues by a factor of two. As a result, routing signals and bias lines across the buses may become difficult if only one or two more layers of metal are available.

If the bias currents drawn from a long bus are relatively well defined, then the bus width can be "tapered" from one end to the other so as to create a relatively constant voltage drop along the line. Illustrated in Fig. 19.47, this technique can be used if the metal resistance and its temperature coefficient are known.

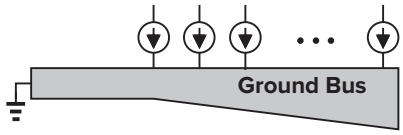


Figure 19.47 Tapered ground line for reduction of voltage drops.

19.2.8 Pads and ESD Protection

The interface between an integrated circuit and the external environment involves a number of important issues. In order to attach bond wires to the die, large "pads" are placed on the perimeter of the chip and connected to the corresponding nodes in the circuit (Fig. 19.48).

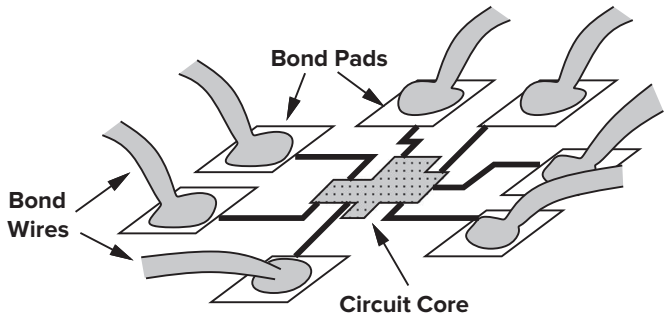


Figure 19.48 Addition of bonding pads to a chip.

The pad dimensions and structure are dictated by the reliability issues and margin for manufacturing tolerances in the wire-bonding process. With bond wire diameters ranging from 25 μm to 50 μm, the minimum pad size falls between roughly 70 μm $\times$ 70 μm and 100 μm $\times$ 100 μm. Adjacent pads are usually separated by at least 25 μm. From the circuit design point of view, the pad dimensions must be minimized so as to reduce both the capacitance of the pad to the substrate and the total die area.

A simple pad would consist of only a square made of the top metal layer. However, such a structure is susceptible to "lift-off" during bonding. For this reason, each pad is typically formed by the two topmost metal layers, connected to each other by many small vias on the perimeter (Fig. 19.49). Note that this structure suffers from a larger capacitance to the substrate than a pad made of only the top layer.

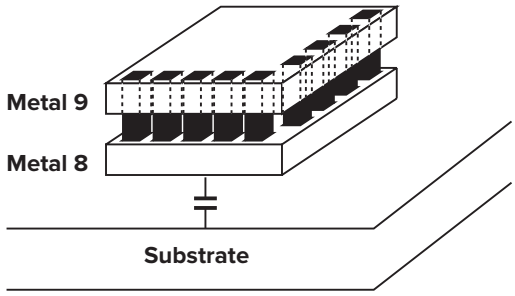


Figure 19.49 Structure of a typical bonding pad.

▶ **Example 19.6**

Calculate the capacitance of a metal 4 pad and a metal 4/metal 3 pad. Assume dimensions of 75 μm × 75 μm and use the capacitance data shown in Fig. 19.50.

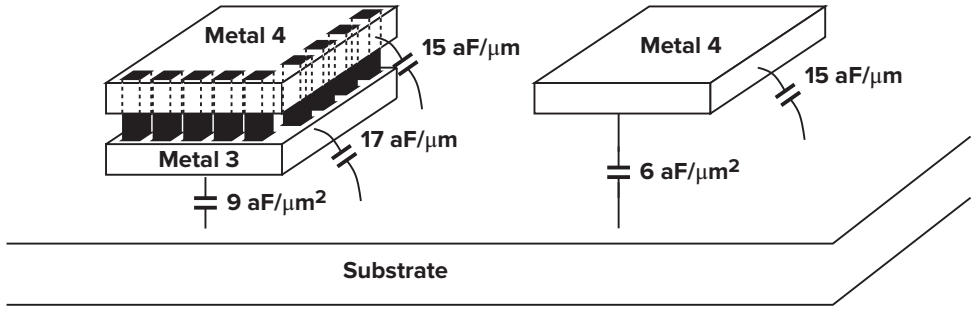


Figure 19.50

Solution

For a metal 4 pad,

$$C_{tot} = 75^2 \times 6 + 75 \times 4 \times 15 \tag{19.15}$$

$$= 38.25 \text{ fF} \tag{19.16}$$

For a metal 4/metal 3 pad,

$$C_{tot} = 75^2 \times 9 + 75 \times 4 \times (17 + 15) \tag{19.17}$$

$$= 60.22 \text{ fF} \tag{19.18}$$

Note that the fringe capacitances of metal 4 and metal 3 are directly added here. This is a rough approximation. ◀

The pads carrying high-frequency signals can be configured as octagons so as to reduce their capacitance. Depicted in Fig. 19.51, such a structure is obtained by removing the corner areas of a square pad—without making the task of bonding more difficult. If $a = b$, then both the area and the perimeter of the pad fall by about 20%.

The interface between an IC and the external world also entails the problem of electrostatic discharge (ESD). This effect occurs when an external object having a high potential touches one of the connections to the circuit. Since the capacitance seen at each input or output is small, the ESD produces a large voltage, damaging the devices fabricated on the chip.

A common case of ESD arises when ICs are handled by human beings. For this effect, the human body can be modeled by a capacitance of a few hundred picofarads in series with a resistance of a few kilohms.

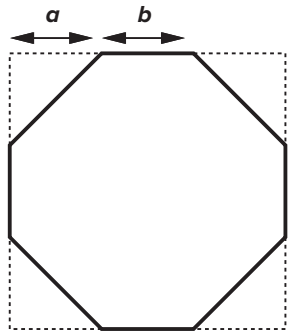


Figure 19.51 Use of octagonal pad to reduce capacitance.

Depending on the environment, the voltage across the capacitance ranges from a few hundred volts to several thousand volts. Thus, if a person touches a line connecting to the chip, the chip is easily damaged. Interestingly, electrostatic discharge may occur even without actual contact because at high electric fields, the person's finger "arcs" to the connection through the air if the finger is sufficiently close to the line.

It is important to note that ESD may occur even without human intervention. If not properly grounded, various objects in a typical chip assembly line accumulate charge, rising to high potential levels. Furthermore, charge in dry air may create substantial potential gradients with respect to ground.

MOS devices sustain two types of permanent damage as a result of ESD. First, the gate oxide may break down if the electric field exceeds roughly 10^7 V/cm (e.g., 2 V for an oxide thickness of 20 $\mathring{\text{A}}$), typically leading to a very low resistance between the gate and the channel. Second, the source/drain junction diodes may melt if they carry a large current in forward or reverse bias, creating a short to the bulk. For today's short-channel devices, both of these phenomena are likely to occur.

In order to alleviate the problem of electrostatic discharge, CMOS circuits incorporate ESD protection devices. Illustrated in Fig. 19.52, such devices clamp the external discharge to ground or V_{DD}, thereby limiting the potential applied to the circuit. Resistor R_1 is usually necessary so as to avoid damaging D_1 or D_2 due to large currents that would otherwise flow from the external source.

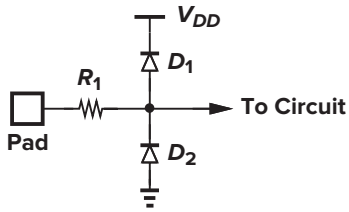


Figure 19.52 Simple ESD protection circuit.

The use of ESD protection structures involves three critical issues. First, the devices introduce substantial capacitances from the node to ground and V_{DD}, degrading the speed and the matching of impedances at the input and output ports of the circuit. Since the protection devices, such as D_1 and D_2 in Fig. 19.52, must be large enough so that the chip sustains a high ESD voltage without damage, their capacitance may reach several picofarads. The thermal noise of R_1 may also become significant.

Second, the parasitic capacitance of the ESD devices may couple noise on V_{DD} to the input of the circuit, corrupting the signal. We return to this issue in Sec. 19.4.

Third, if not properly designed, ESD structures may lead to latch-up in CMOS circuits when electrostatic discharge occurs during actual circuit operation (or even when the circuit is turned on). For this reason, process engineers fabricate and characterize many different ESD structures for each generation of a technology, eventually providing a few reliable configurations that can be used in circuits.[6]

[6]In general, a circuit designer should not use an ESD structure that has not been tested and qualified for the technology. Uncharacterized ESD devices are likely to cause latch-up.

19.3 ■ Substrate Coupling

Most modern CMOS technologies use a heavily-doped p^+ substrate to minimize latch-up susceptibility. However, the low resistivity of the substrate (on the order of 0.1 Ω·cm) creates unwanted paths between various devices in the circuit, thereby corrupting sensitive signals. Called "substrate coupling" or "substrate noise," this effect has become a serious issue in today's mixed-signal ICs [2].

To understand this phenomenon, suppose a CMOS inverter sensing a clock is laid out next to a common-source stage amplifying an analog signal [Fig. 19.53(a)]. Note that the substrate is connected

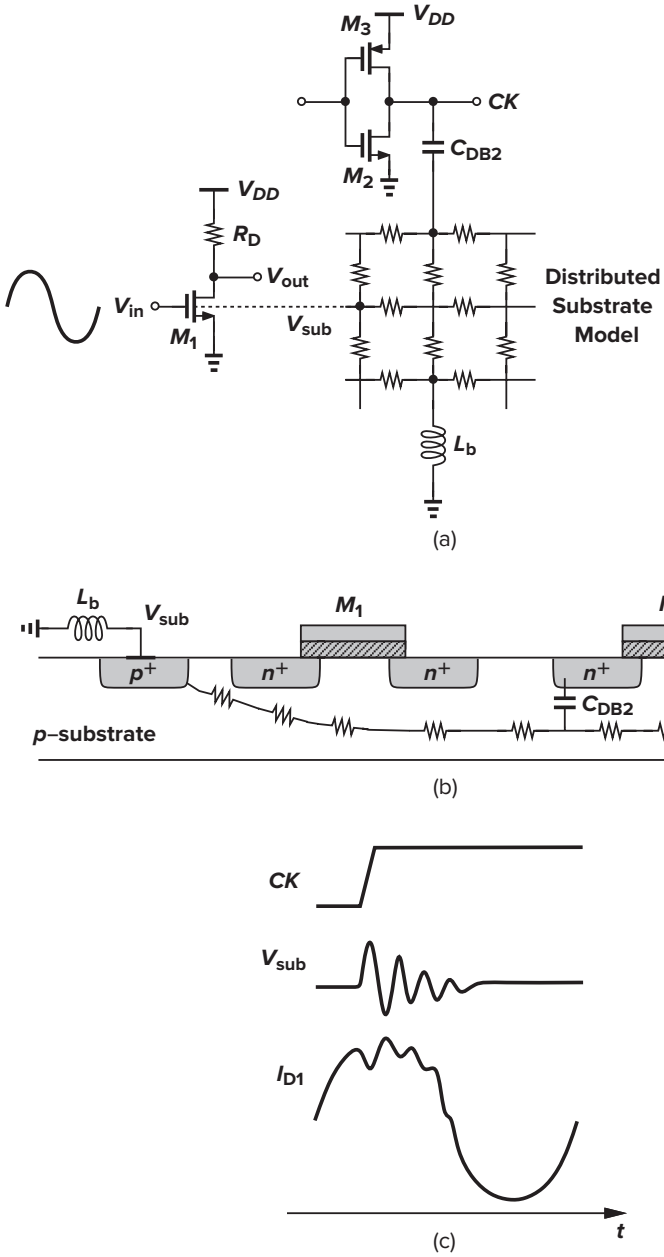


Figure 19.53 (a) Mixed-signal circuit including the effect of substrate coupling; (b) side view of device layout; (c) signal waveforms.

to ground through a bond wire that exhibits an (unwanted) inductance of L_b. With the aid of the cross section depicted in Fig. 19.53(b), we observe that the large voltage excursions at the drain of M_2 are coupled to the substrate through the drain junction capacitance, disturbing the substrate voltage because of the finite impedance of L_b.

How does the substrate noise influence M_1? The principal coupling mechanism here occurs through body effect, varying the threshold voltage of M_1 with the substrate voltage. Since the drain current of M_1 depends on $V_{in} - V_{TH1}$, variations in V_{TH1} are indistinguishable from those in V_{in}. In other words, as illustrated in Fig. 19.53(c), every transition of CK disturbs the analog output.

The problem of substrate coupling becomes more noticeable as the number of "noise" generators increases. In a mixed-signal environment, thousands of digital gates may inject noise into the substrate—especially during clock transitions—introducing hundreds of millivolts of disturbance in the substrate potential. The disturbance is also proportional to the size of the noise-injecting devices, an important issue if large transistors are used as buffers driving heavy external loads.

It may seem that substrate coupling can be decreased by increasing the physical spacing between sensitive building blocks and digital sections of a chip. In practice, however, this remedy may not be effective or feasible. If heavily doped, the substrate operates as a low-resistance plane, distributing a relatively uniform potential across the chip regardless of the position of the noise generators [3]. Furthermore, in many mixed-signal systems, the analog and digital functions are so heavily blended that it is difficult to separate their corresponding circuits. Figure 19.54 shows a slice of an A/D converter consisting of a comparator, a flipflop, a NAND gate, and a read-only memory (ROM). Various logical swings in the comparator and the digital circuits generate substrate noise, but increasing the distance between any two blocks necessitates long interconnects, degrading the performance.

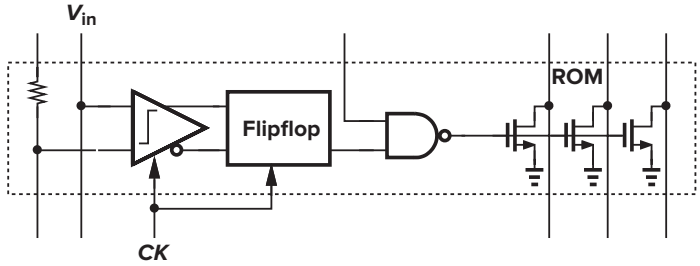


Figure 19.54 A slice of an A/D converter.

In order to minimize the effect of substrate noise, the following methods can be applied. First, differential operation should be used throughout the circuit, making the analog section less sensitive to common-mode noise. Second, digital signals and clocks should be distributed in complementary form, thereby reducing the net amount of the coupled noise. Third, critical operations, e.g., sampling a signal or transferring charge from one capacitance to another, should be performed well after clock transitions so that the substrate voltage settles. Fourth, the inductance of the bond wire connected to the substrate should be minimized (Sec. 19.4). Also, op amps using a PMOS differential input are preferred because the well of the transistors can be tied to their common source, reducing the effect of substrate noise.

In circuits fabricated on lightly-doped substrates, "guard rings" can be employed to isolate the sensitive sections from the substrate noise produced by other sections. A guard ring may be simply a continuous ring made of substrate ties that surrounds the circuit, providing a low-impedance path to ground for the charge carriers produced in the substrate. With its large depth, the n-well can also augment the operation of a guard ring by stopping the noise currents flowing near the surface (Fig. 19.55).

In large mixed-signal ICs, it may not be possible to avoid substrate "bounce" with respect to the external ground because of the high transient currents drawn by the devices and the finite impedance of

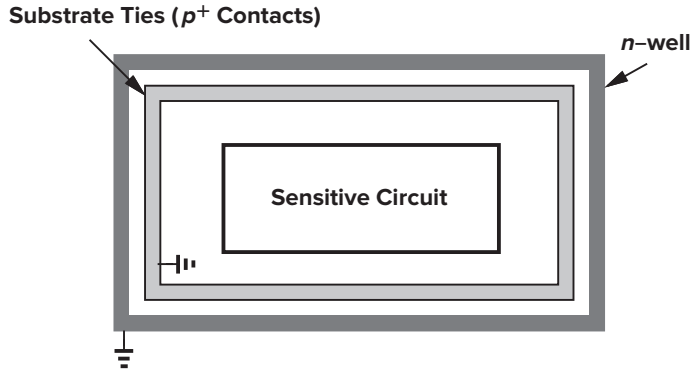


Figure 19.55 Use of guard ring to protect sensitive circuits.

the bond wire connected to the substrate. However, we recognize that if the ground of the chip bounces in unison with the substrate, then the transistors experience no noise. Illustrated in Fig. 19.56, this idea suggests that the ground and the substrate should be connected on the chip and brought out through a single wire.

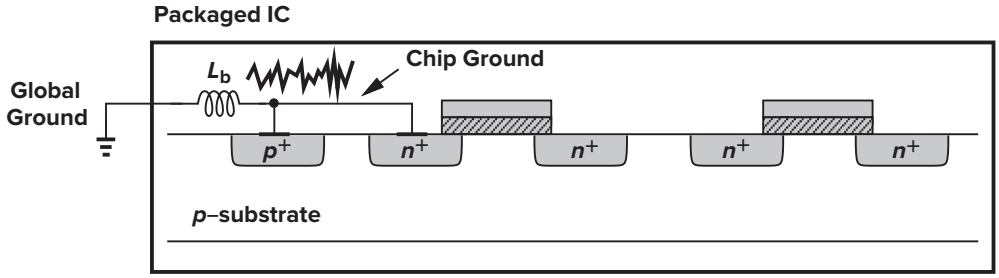


Figure 19.56 Substrate bounce.

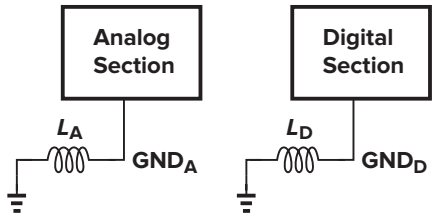


Figure 19.57 Analog and digital grounds.

The connection of the substrate to the chip ground nonetheless faces two difficulties. The first relates to "ground bounce." As shown in Fig. 19.57 and explained in Sec. 19.4, most mixed-signal circuits employ at least one "analog ground" and one "digital ground" so as to avoid corrupting the analog section by the large transient noise produced by the digital section. To which ground should the substrate be connected? If the analog ground is used, then the large substrate noise current must flow through L_A, creating noise on GND_A [Fig. 19.58(a)], and if the digital ground is used, then the substrate voltage is heavily disturbed by the large noise on GND_D [Fig. 19.58(b)]. Of course, connecting the substrate to both GND_A and GND_D gives rise to a low-resistance path between the two, defeating the purpose of separating the analog and digital grounds.

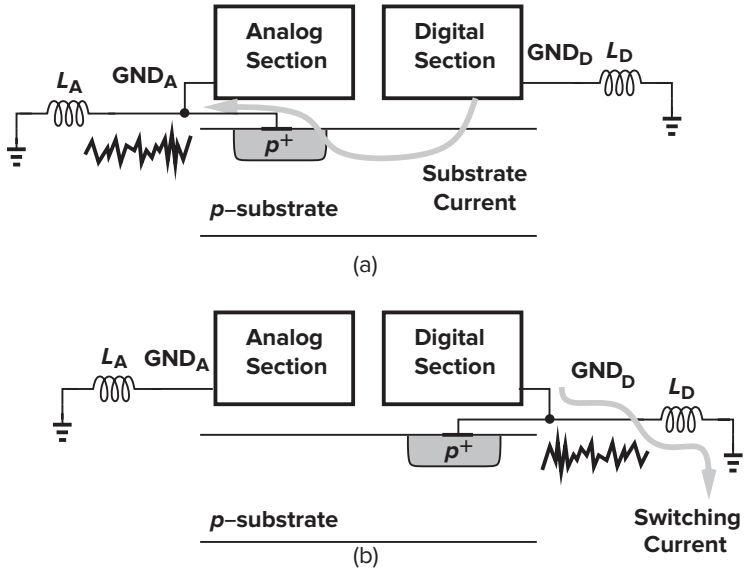


Figure 19.58 Connection of substrate contact to (a) analog ground and (b) digital ground.

The choice between the configurations shown in Figs. 19.58(a) and (b) depends on the transient currents drawn by the digital section from the substrate and the ground as well as the magnitudes of L_A and L_D. In most cases, the topology of Fig. 19.58(a) is preferred because it ensures that the analog ground voltage and the substrate potential vary in unison. As illustrated in Fig. 19.59(a), if the analog ground and the substrate experience unequal bounce, then the drain current of M_1 is corrupted by the substrate noise. The configuration of Fig. 19.59(b), on the other hand, introduces less noise in I_{D1}. In general, careful, realistic simulations of the overall environment (including the package) are necessary to determine which approach yields less noise.

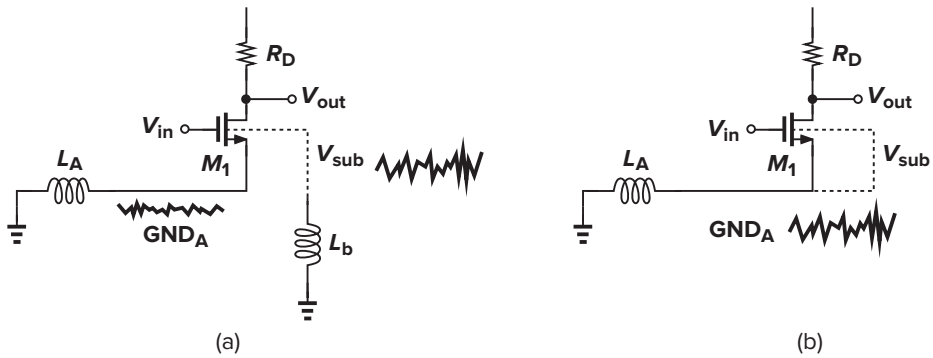


Figure 19.59 (a) Large source-bulk noise voltage due to separating substrate contact from analog ground; (b) suppression of the effect.

The second issue in allowing the substrate and a chip ground to bounce together is the difficulty in defining a reference potential for the input signals. As shown in Fig. 19.60(a), a single-ended input is heavily corrupted as its reference point changes from the off-chip ground to the on-chip ground. That is, $V'_{in} \neq V_{in}$, even though the substrate and the ground bounce in unison. For the differential structure of Fig. 19.60(b), the effect is much less pronounced, but in high-precision applications, asymmetries in the circuit and interconnections convert a fraction of the common-mode noise to a differential component.

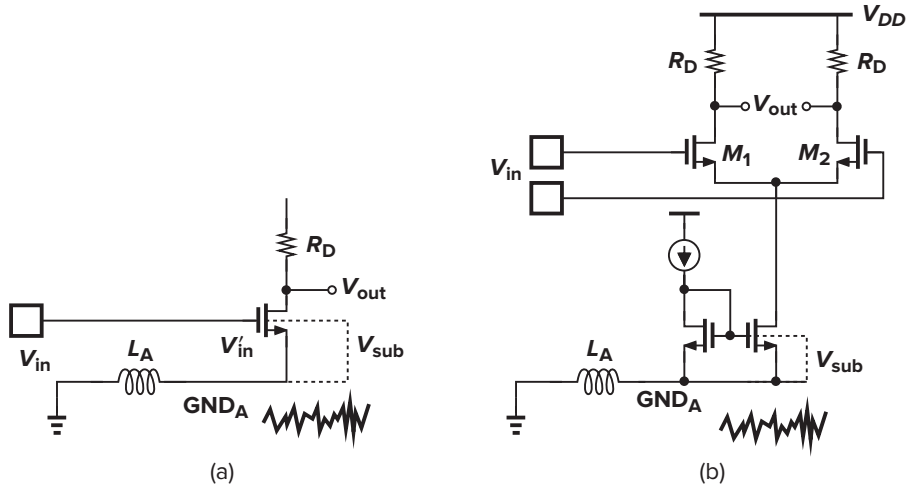


Figure 19.60 (a) Input signal corruption due to ground and substrate bounce; (b) less corruption in a differential environment.

19.4 ■ Packaging

After fabrication and dicing, integrated circuits are packaged. The parasitics associated with the package and connections to the chip introduce many difficulties in the evaluation of the actual performance of the circuit at high speeds and/or high accuracies.

Let us first consider a simple dual-in-line package (DIP) [Fig. 19.61(a)]. Here, the die is mounted in the center cavity and bonded to the pads on the perimeter of the cavity. These pads are in fact the tip of each trace that ends in each package pin. Such a structure exhibits the following parasitics: bond wire self-inductance, trace self-inductance, trace-to-ground capacitance, trace-to-trace mutual inductance, and trace-to-trace capacitance. Thus, as shown in Fig. 19.61(b), the connections between the circuit and the external world are far from ideal.

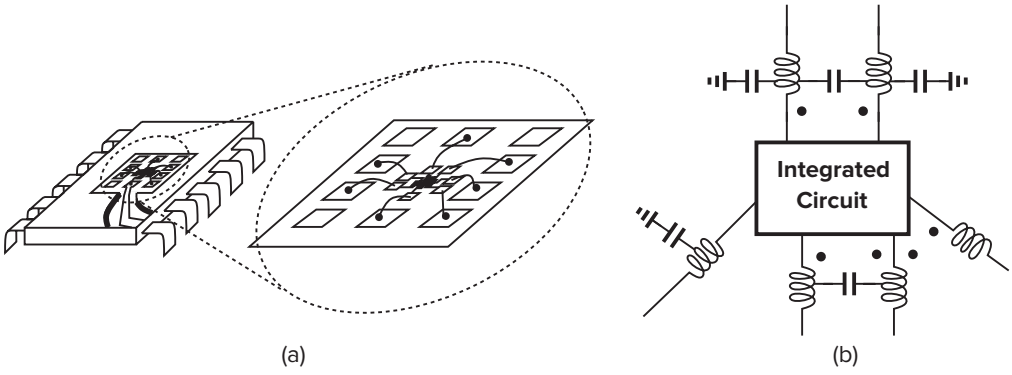


Figure 19.61 (a) Dual-in-line package; (b) electrical model of the package.

While, owing to both circuit innovations and device scaling, the speed and accuracy of integrated circuits have steadily increased, the performance of packages, especially for low-cost applications, has not improved significantly. This limitation originates from the unscalable nature of packages and the environment in which they are used. For example, the diameter of the bond wires, the width and spacing

of package pins, and the width and spacing of the traces in printed circuit (PC) boards are determined by mechanical stress, ease and cost of assembly, series resistance at high frequencies (skin effect), etc. In the past 20 years, these dimensions have scaled by less than a factor of five, whereas the speed of many mixed-signal circuits has increased by two orders of magnitude. As a result, packaging continues to limit the achievable performance of today's high-performance ICs.

The foregoing issues dictate that the package parasitics be taken into account in the design of integrated circuits—sometimes from the very beginning. Thus, simulations must include a reasonable circuit model of the package, and the design and layout must take many measures to minimize the effect of package parasitics.

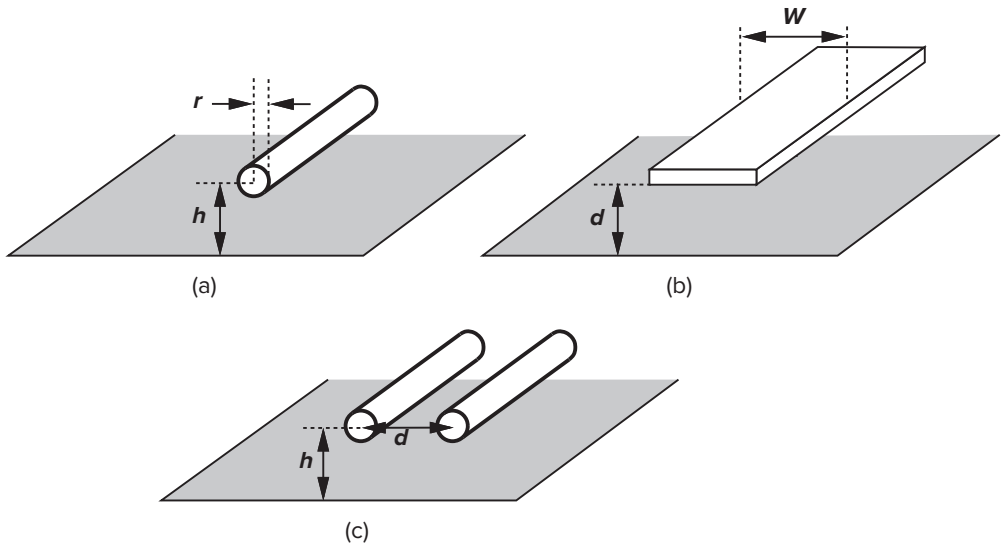


Figure 19.62 Common geometries in packaging.

Since some package manufacturers do not provide circuit models for their products, IC designers often develop the models themselves by calculations and measurements. Figure 19.62 depicts three common cases of self- and mutual inductance. From [6], we have for a round wire above a ground plane [Fig. 19.62(a)]

$$L \approx 0.2 \ln \frac{2h}{r} \text{ nH/mm} \tag{19.19}$$

which amounts to roughly 1 nH/mm for typical bond wires. For a flat trace above a ground plane [Fig. 19.62(b)],

$$L \approx \frac{1.6}{K_f} \cdot \frac{d}{W} \text{ nH/mm} \tag{19.20}$$

where K_f denotes the fringe factor and from the data in [6] can be approximated as $0.72(d/W) + 1$. For two round wires above a ground plane, the mutual inductance is [6]

$$L_m = 0.1 \ln \left[1 + \left(\frac{2h}{d} \right)^2 \right] \text{ nH/mm} \tag{19.21}$$

The parasitic capacitances can be calculated with the simple interplate equation and Eq. (18.11). In practice, the bond wires are not simply straight, parallel lines, requiring electromagnetic field simulations for proper modeling.

Let us now study the effect of each type of package parasitic. We categorize the connections to the chip into five groups: power and ground lines, analog and clock inputs, outputs, reference lines, and substrate connection(s).

Self-Inductance Each bond wire and its corresponding package trace exhibit a finite self-inductance, with a total value between approximately 2 nH and 20 nH depending on the length of the wire and the type of the package. To understand how the self-inductance of supply and ground lines affects the performance, suppose a mixed-signal circuit incorporates a CMOS inverter as a clock buffer to drive a moderate on-chip capacitance, e.g., 0.5 pF (Fig. 19.63). Also, assume that the buffered clock must have transition times less than 0.5 ns, thereby demanding a current of $C\Delta V/\Delta t = 3$ mA. Since this current is drawn from V_{DD1} and GND_1 in 0.5 ns, we can estimate the voltage drop across L_D or L_G as[7] $L\Delta I/\Delta t = 6 \times 10^6 L$. For example, if $L_D = L_G = 5$ nH, then the transient voltage across each inductor equals 30 mV. This effect is called supply and ground "bounce" or "noise." Note that if the inverter is replaced by a differential pair, the supply bounce decreases substantially (why?), another advantange of differential operation.

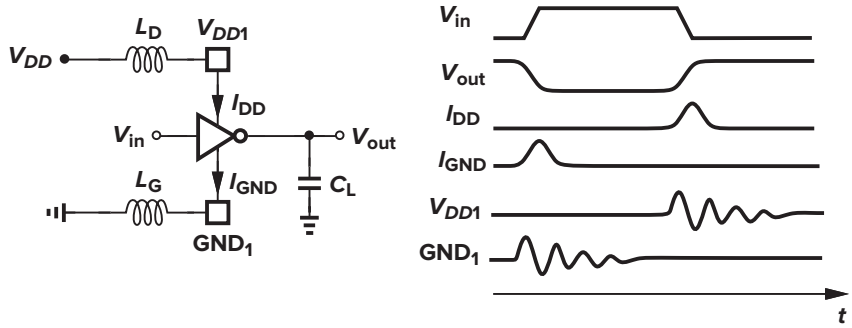


Figure 19.63 CMOS inverter driving a load capacitance.

A supply noise of 30 mV may seem quite benign, especially if the analog circuits feeding from the same supply line are fully differential. However, in a typical mixed-signal IC, hundreds or thousands of digital gates may switch during each clock transition, creating enormous noise on their supply and ground connections. For this reason, most such systems employ separate supply and ground lines for the analog and digital sections; hence the terminology "analog supply" and "digital supply."

Separating power lines into analog and digital groups is not always straightforward. As an example, suppose a sampling circuit is clocked by an inverter (Fig. 19.64). Should the inverter be supplied from analog or digital power lines? If the inverter is connected to the digital supply, then the large noise on V_{DD} couples through the gate-drain overlap capacitance of M_1, corrupting V_{out} when the transistor is off. On the other hand, if many such inverters are supplied from the analog V_{DD}, they collectively draw large transient currents, corrupting the supply voltage. These cases may require a third type of power line so that it remains less noisy than the digital supplies.

For characterization and troubleshooting purposes, it is sometimes desirable to monitor the supply noise. Figure 19.65 illustrates a simple method whereby a PMOS device sensing the noise between the on-chip supply and ground lines injects a current into an external 50-Ω transmission line and measurement

[7]This calculation is quite rough because the current produced by the buffer varies during the transition.

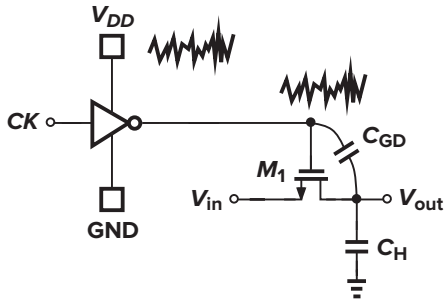


Figure 19.64 Noise in a sampling circuit resulting from the clock buffer's supply bounce.

apparatus [2]. Since the transconductance of M_1 can be determined by a small, static change in V_{DD}, the measurement readily reveals both the magnitude and the shape of the supply noise.

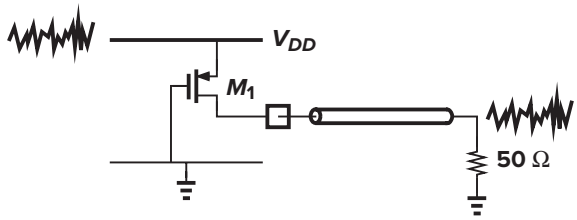


Figure 19.65 Measurement of supply noise.

In cases where a single connection to the chip sustains a prohibitively large transient voltage (e.g., if in Fig. 19.63 or 19.64 many inverters switch simultaneously), multiple pads, bond wires, and package pins are used so as to reduce the equivalent inductance (Fig. 19.66).

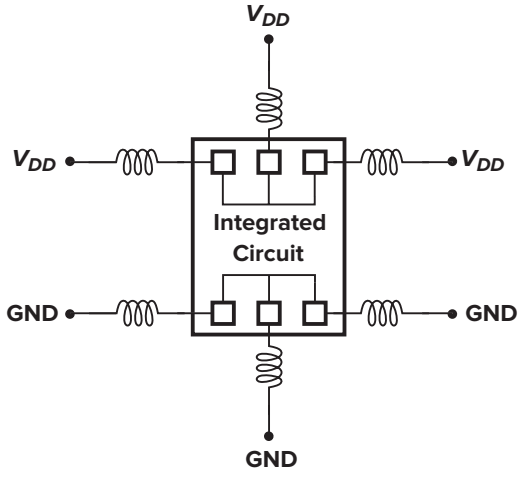


Figure 19.66 Use of multiple wires to reduce overall inductance.

▶ **Example 19.7**

In a 600-MHz, 2-V CMOS microprocessor containing 15 million transistors, the supply current varies by 25 A in approximately 5 ns [5]. If the processor provides 200 bond wires for ground and 200 for V_{DD}, estimate the resulting supply bounce.

Solution

Assuming a total inductance of 5 nH for each bond wire and its corresponding package trace and pin, we have

$$\Delta V = L\frac{\Delta I}{\Delta t} \tag{19.22}$$

$$= \frac{5 \times 10^{-9}}{200} \cdot \frac{25}{5 \times 10^{-9}} \tag{19.23}$$

$$= 125 \text{ mV} \tag{19.24}$$

In the worst case, the supply bounce and the ground bounce add in-phase, yielding a total noise of roughly 250 mV, greater than 10% of the nominal supply voltage. To further suppress the noise, an external 1-μF MOS capacitor is placed on top of the chip and another 160 supply and ground bond wire pairs are connected from the chip to the capacitor [5].

◀

In some applications, high transient currents drawn from the supply make it difficult to maintain a small bounce on the supply and ground individually. In such cases, a large on-chip capacitor may be used to stabilize the *difference* between V_{DD} and ground. Illustrated in Fig. 19.67, the idea is that if C_1 is sufficiently large, then V_{DD1} and GND_1 bounce in unison. As mentioned earlier, the residual noise on GND_1 may be negligible if the input signals are differential.

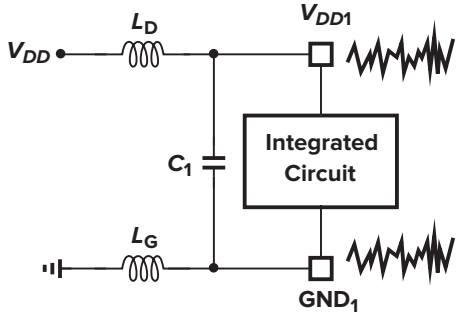


Figure 19.67 On-chip capacitor used to lower supply-ground noise voltage.

This remedy nonetheless involves several issues. First, the value of the capacitor must be chosen carefully because it may otherwise *resonate* with the package inductance at the operating frequency of the chip (e.g., the clock frequency or its harmonics or subharmonics), thereby *amplifying* the supply and ground noise. For this reason, some resistance is added in series with the capacitor (or a MOS capacitor is sized such that its channel resistance dampens the resonance) [5]. Even in the absence of exact resonance, an insufficient value of the decoupling capacitor may simply give rise to slower ringing on the power lines. Second, since the capacitor is usually formed by a very large MOS transistor (actually, as explained in Sec. 18.7.2, a large number of MOSFETs in parallel), the yield of the circuit may suffer. This is because, for the capacitor to be effective, its total area is typically comparable with the total gate area of all of the transistors in the circuit; e.g., it is as if the number of transistors on the chip were doubled.

Self-inductance also manifests itself in the connection to the substrate. As mentioned in Sec. 19.3, with the large transient currents injected by the devices into the substrate, a low-impedance connection is necessary to minimize the substrate bounce. As shown in Fig. 19.68, some modern packages contain a metal ground plane to which the die can be attached by conductive epoxy. The plane ends in several package pins that are tied to the board ground. Avoiding bond wires and long, narrow traces in the substrate connection, such packages substantially reduce the substrate noise with no additional assembly cost. In more expensive packages, the ground plane is exposed on the bottom and can be directly attached

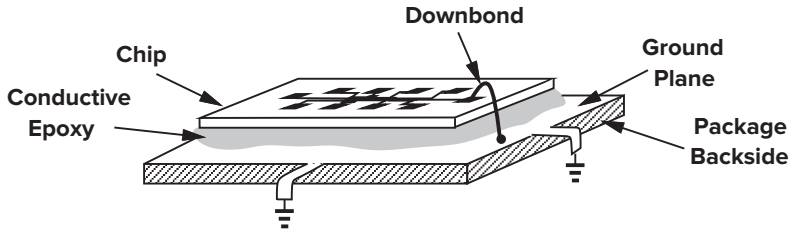


Figure 19.68 Package using a ground plane for substrate connection.

to the board ground, thus avoiding the inductance of the package pins. Also, the ground pads of the circuit can be "downbonded" to the underlying plane to minimize their inductance (while increasing the cost).

The effect of self-inductance must also be considered for input signals. The inductance, along with the pad capacitance and the circuit's input capacitance, forms a low-pass filter, attenuating high-frequency components and/or creating severe ringing in transient waveforms. For example, in the precision multiply-by-two circuit described in Sec. 13.3.3, when the two capacitors are switched to the input, the package inductance may limit the settling speed.

Some ICs require constant voltages that must be provided externally. Such voltages may serve as an accurate reference, e.g., in A/D or D/A converters, or to define some bias points on the chip. The package inductance degrades the settling behavior if the circuit injects significant switching noise into the reference.

▶ Example 19.8

Differential pairs are often used as "current switches." As shown in Fig. 19.69, the circuit routes its tail current to either of the outputs according to the large swings controlling the gates of M_1 and M_2. Explain what happens at node X during switching. If the tail currents of a large number of differential pairs feed from node X, should this voltage be provided externally?

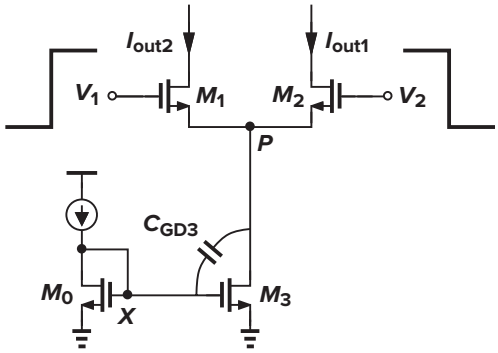


Figure 19.69 Differential pair operating as a current switch.

Solution

Recall from Chapter 4 that for the differential pair to experience complete switching, the differential swing $|V_2 - V_1|$ must exceed $\sqrt{2}(V_{GS} - V_{TH})_{eq}$, where $(V_{GS} - V_{TH})_{eq}$ is the overdrive of M_1 and M_2 in equilibrium, i.e., if $I_{D1} = I_{D2}$. We denote the voltage at node P when the pair is completely switched by V_{P1}, and in equilibrium by V_{P2}. Thus,

$$V_{P1} = V_2 - \sqrt{2}(V_{GS} - V_{TH})_{eq} \tag{19.25}$$

In equilibrium,

$$V_{P2} = \frac{V_1 + V_2}{2} - (V_{GS} - V_{TH})_{eq} \tag{19.26}$$

Assuming that $V_2 - V_1 = \sqrt{2}(V_{GS} - V_{TH})_{eq}$, and hence $V_1 = V_2 - \sqrt{2}(V_{GS} - V_{TH})_{eq}$, we have

$$V_{P2} = V_2 - \left(1 + \frac{\sqrt{2}}{2}\right)(V_{GS} - V_{TH})_{eq} \tag{19.27}$$

Thus, V_{P2} is *lower* than V_{P1} by $(1 - \sqrt{2}/2)(V_{GS} - V_{TH})_{eq}$, indicating that during switching, V_P drops by this amount. This voltage change is coupled to node X through the gate-drain overlap capacitance of M_3, disturbing I_{D3} and hence I_{out1} or I_{out2}.

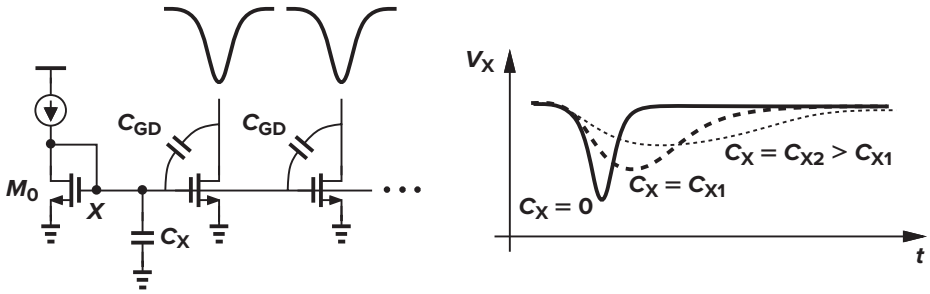


Figure 19.70 Addition of on-chip bypass capacitor to suppress noise at node X.

With a large number of current switches connected to node X, the disturbance may be quite significant, demanding that a decoupling capacitor be connected from node X to ground (Fig. 19.70). However, such a capacitor along with the small-signal resistance of M_0 introduces a long settling time at node X, possibly degrading the overall speed. To avoid this effect, C_X may need to be 100 to 1,000 times the *total* gate-drain overlap capacitance that injects noise into X. If such a large capacitor is placed off-chip, it actually appears in series with the package inductance (Fig. 19.71). In general, careful simulations are necessary to determine the preferable choice here. In many cases, leaving node X *agile* yields the fastest settling.

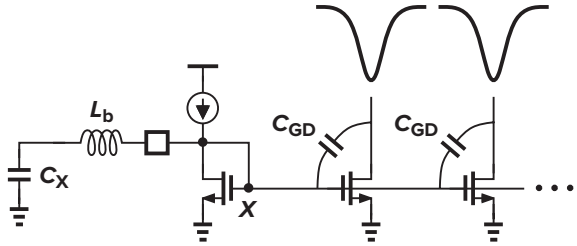


Figure 19.71 Addition of bypass capacitor externally.

The self-inductance of package connections also affects the performance of digital output buffers. In high-speed systems, these drivers must deliver tens of milliamps of current to the load with fast transitions. With many such buffers operating in a mixed-signal circuit, the resulting voltage drops on the power lines may become very large, increasing the rise time and fall time of the digital outputs and corrupting their timing.

Mutual Inductance While dedicating separate power lines to analog and digital sections reduces the noise on the analog supply, some noise may still couple to sensitive signals through the mutual inductance of bond wires and package traces. As illustrated in Fig. 19.72, both analog supplies and analog inputs are susceptible to noise or transitions on digital supplies, clock lines, or output buffers. With an arbitrary pad configuration, even differential signaling cannot eliminate this effect because the noisy lines may not surround the sensitive lines symmetrically. Thus, the design of the pad frame and the position of the pads play a critical role in the performance that can be achieved.

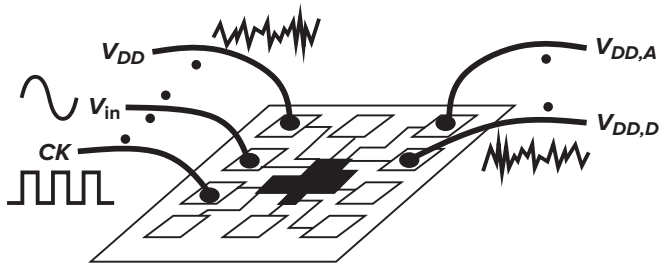


Figure 19.72 Coupling due to mutual inductance between wires.

Mutual inductance also manifests itself in parallel bond wires used to lower the overall self-inductance of a connection (Fig. 19.73). For two such wires, the equivalent inductance is equal to $(L_S + M)/2$, where M denotes the mutual inductance, rather than $L_S/2$.

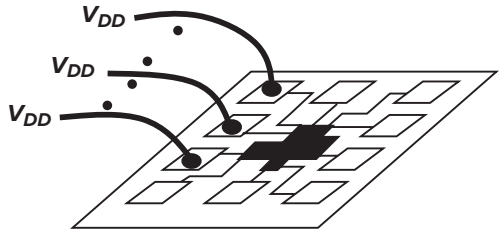


Figure 19.73 Multiple supply bond wires with mutual coupling.

Two methods can reduce the mutual coupling between inductors. First, the wires can be connected such that they are perpendicular to each other, i.e., they terminate on perpendicular sides of the chip [Fig. 19.74(a)]. Second, (quiet) ground or supply lines can be interposed between critical bond wires [Fig. 19.74(b)]. As shown in Fig. 19.74(c), even if several parallel lines are surrounded by ground wires, the effect of mutual inductance drops to negligible values.

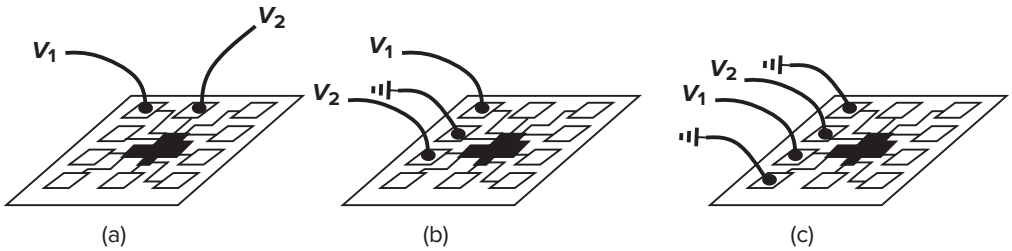


Figure 19.74 Reduction of mutual coupling by (a) perpendicular lines, (b) additional ground lines, and (c) occasional ground lines.

It is also interesting to note that mutual inductance *reduces* the self-inductance of two wires if they carry currents in opposite directions. If, as shown in Fig. 19.75, the supply and ground lines of a circuit

are in parallel, then the total inductance equals $2L_S - M$ rather than $2L_S$. This observation proves useful in designing the pad frame and determining the package connections.

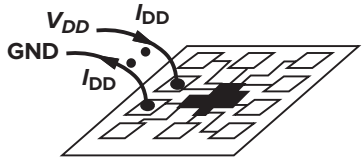


Figure 19.75 Reduction of mutual inductance between two wires carrying equal and opposite currents.

Self- and Mutual Capacitance The capacitance seen from each trace of the package to ground may limit the input bandwidth of the circuit or load the preceding stage. More important, this capacitance and the total inductance of the bond wire and the package trace yield a finite resonance frequency that may be stimulated by various transient currents drawn by the circuit. Since the wires and traces exhibit a small series resistance, a high quality factor (Q) results, giving rise to a sharp resonance and amplifying the noise considerably. The capacitance between the traces leads to additional coupling between lines and must be included in simulations.

References

[1] N. C. C. Lu et al., "Modeling and Optimization of Monolithic Polycrystalline Silicon Resistors," *IEEE Trans. Electron Devices*, vol. ED-28, pp. 818–830, July 1981.

[2] D. Su et al., "Experimental Results and Modeling Techniques for Substrate Noise in Mixed-Signal Integrated Cicuits," *IEEE J. of Solid-State Circuits*, vol. 28, pp. 420–430, April 1993.

[3] T. Blalack and B. A. Wooley, "The Effects of Switching Noise on an Oversampling A/D Converter," *ISSCC Dig. of Tech. Papers*, pp. 200–201, February 1995.

[4] B. Razavi, *Principles of Data Conversion System Design* (New York: IEEE Press, 1995).

[5] D. W. Dobberpuhl, "Circuits and Technology for Digital's StrongARM and ALPHA Microprocessors," *Proc. of 17th Conference on Advanced Research in VLSI*, pp. 2–11, September 1997.

[6] N. K. Verghese, T. J. Schmerbeck, and D. J. Allstot, *Simulation Techniques and Solutions for Mixed-Signal Coupling in Integrated Circuits* (Boston: Kluwer Academic Publishers, 1995).

Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume that $V_{DD} = 3$ V where necessary. Also, assume that all transistors are in saturation.

19.1. In Fig. 19.3, polysilicon has a sheet resistance of 30 Ω/□ (before silicidation) and metal 1 a sheet resistance of 80 mΩ/□. What is the ratio of the resistivities of the two materials?

19.2. A MOSFET with $W/L = 100\ \mu$m$/0.5\ \mu$m undergoes ideal scaling by a factor of two. What happens to the sheet resistivity and the total resistance of the gate?

19.3. A cascode structure uses $W/L = 100\ \mu$m$/0.5\ \mu$m for both the input device and the cascode device. If the sheet resistance of polysilicon is 5 Ω/□ and the maximum tolerable gate resistance 10 Ω, draw the layout of the structure while minimizing the drain junction capacitances.

19.4. In Fig. 19.7, explain what happens to the differential amplifier if each of the design rules A_1–A_8 is violated.

19.5. The input differential pair of an amplifier is to be laid out as in Fig. 19.19, but with each half device (e.g., $1/2M_1$) using four gate fingers. What is the minimum number of interconnect layers required here?

19.6. Large integrated circuits may suffer from significant temperature gradients. Compare the performance of the circuits shown in Figs. 19.23 and 19.24 in such an environment.

19.7. Suppose polysilicon with silicide block has a sheet resistance of 60 $\Omega/\square$ and a parallel-plate capacitance of 100 aF/μm^2 to the substrate. Also, assume that these parameters are respectively equal to 2 k$\Omega/\square$ and 1,000 aF/μm^2 for the n-well. Determine which material should be used to construct a 500-Ω resistor if matching considerations require a minimum poly width of 3 μm and a minimum n-well length of 6 μm. Neglect fringe capacitances.

19.8. Using the data in Table 18.1, calculate C and C_P for each structure in Fig. 19.35 and identify the one with minimum C_P/C. Neglect fringe capacitances.

19.9. A metal 4 wire with a length of 1,000 μm and width of 1 μm is driven by a source impedance of 500 Ω. Using the data in Table 18.1 and assuming a sheet resistance of 40 m$\Omega/\square$, calculate the delay through the wire and compare the result with the lumped time constant obtained by multiplying the source impedance by the total wire capacitance.

19.10. Repeat Problem 19.9 if the width of the wire is increased to 2 μm.

19.11. An interconnect having a length of 1,000 μm is required in a circuit. Using the data in Table 15.1 and assuming that the sheet resistance of metals 1–3 is 80 m$\Omega/\square$ and that of metal 4 is 40 m$\Omega/\square$, determine which metal layer must be used to obtain the minimum delay.

19.12. Some new technologies use copper for interconnects because its resistivity is about half that of aluminum. Repeat Problem 19.11 with copper interconnects.

19.13. In the circuit of Fig. 19.53(a), $(W/L)_1 = 100/0.5$ and $I_{D1} = 1$ mA. If the substrate noise, V_{sub}, has a peak-to-peak amplitude of 50 mV, what is the effect referred to the gate of M_1?

19.14. Suppose two bond wires are placed 5 mm above ground with a center-to-center spacing of 1 mm.

(a) What is the total mutual inductance if each wire is 4 mm long?

(b) If one wire carries a 100-MHz sinusoidal current with a peak amplitude of 1 mA, what is the voltage induced across the other wire?

19.15. In Problem 14, what center-to-center spacing is required to decrease the induced voltage by a factor of four?

19.16. In order to reduce the total bond wire inductance, a package uses 4 supply pads and 4 ground pads. Suppose the self-inductance of each wire is 4 nH and the mutual inductance between adjacent lines is 2 nH. Neglecting mutual inductance between nonadjacent lines, calculate the equivalent inductance of the supply and ground connections if (a) all of the supply wires are placed next to each other and so are the ground wires, and (b) every supply wire is placed next to a ground wire.

19.17. The input bandwidth of high-speed circuits may be limited by the bond wire inductance and the pad capacitance. Consider two cases: (a) the bond wire diameter is 50 μm and the pad size 100 μm $\times$ 100 μm; (b) the bond wire diameter is 25 μm and the pad size 50 μm $\times$ 50 μm. If all other dimensions are constant, which case is preferable?